

COMPREHENSIVE INVESTIGATION OF A SEXTORTION EMAIL CAMPAIGN

Infrastructure, Behavioral, and Blockchain Analysis of a Coordinated
Cryptocurrency-Based Extortion Operation



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Executive Summary

This report presents a comprehensive investigation into a sextortion email campaign identified across five distinct email samples. The campaign leverages spoofed email infrastructure and psychological coercion to extort victims into transferring Bitcoin payments.

Through in-depth analysis of email content, authentication mechanisms, infrastructure components, and blockchain transactions, the campaign is assessed as coordinated, scalable, and financially motivated, with no evidence of actual system compromise.

The attacker employs a distributed infrastructure model, combining cloud-based resources, domain manipulation techniques, and a structured cryptocurrency transaction pipeline. Despite rotating technical indicators, strong behavioral consistency enables correlation across all samples.

The findings confirm that the threat is behavioral rather than technical, highlighting the importance of intelligence-driven detection focused on adversarial patterns and tactics.

The analysis further identified structured cryptocurrency activity across multiple wallet chains, including reuse of intermediary wallets across different email samples. Transaction patterns consistently demonstrate multi-input aggregation, rapid fund forwarding, and absence of balance retention, confirming the presence of a coordinated and reusable financial processing pipeline.

1. Introduction

Sextortion campaigns have emerged as a prevalent form of cyber-enabled financial fraud, characterized by the use of large-scale email distribution and psychological coercion rather than technical intrusion. In these campaigns, threat actors send deceptive messages claiming to have compromised the victim's device, obtained sensitive data, or recorded compromising material. Victims are then pressured to transfer funds—most commonly in cryptocurrency—under the threat of public exposure or data misuse.

Unlike traditional cyberattacks that rely on exploiting vulnerabilities or deploying malware, sextortion campaigns operate primarily through **social engineering**, leveraging fear, urgency, and perceived credibility to manipulate recipients into compliance. This approach allows attackers to scale operations efficiently while minimizing technical complexity and risk.

This report presents a comprehensive investigation into a sextortion campaign identified across five distinct email samples; each associated with a separate Bitcoin wallet. While individual technical indicators such as IP addresses, domains, and wallet identifiers vary between samples, the campaign exhibits a high degree of consistency in structure, delivery mechanisms, and financial behavior. These similarities strongly indicate an automated and coordinated operation rather than isolated incidents.

The investigation focuses on analyzing email content and authentication mechanisms, identifying, and assessing the underlying sending infrastructure, and tracing cryptocurrency transactions to uncover patterns of financial movement and operational linkage. By correlating findings across all samples, the report aims to reconstruct the campaign's operational model and evaluate its tactics, techniques, and procedures.

To support this analysis, the report incorporates established threat intelligence frameworks, including MITRE ATT&CK, Cyber Kill Chain, and the Diamond Model of Intrusion Analysis. These frameworks provide a structured methodology for mapping observed behaviors to known adversarial patterns and for translating technical findings into actionable intelligence.

Ultimately, this investigation demonstrates how combining email forensics, infrastructure analysis, blockchain tracing, and intelligence modeling enables a comprehensive understanding of modern financially motivated cyber threats, even in the absence of direct system compromise.

2. Scope and Methodology

- Analysis of five email samples
- Examination of associated infrastructure (IPs, domains)
- Analysis of five Bitcoin wallets and transaction flows
- Cross-correlation of all findings

Methodology:

- Email header forensics
- Infrastructure intelligence analysis
- Blockchain tracing
- Behavioral correlation
- Threat modeling

3. Email Content Analysis

All emails follow a consistent template:

- Claims of device compromise
- Threats of data exposure
- Fixed Bitcoin payment demand
- Strict deadline

No personalization beyond the recipient email address is observed, as shown in Figure 1.

Assessment:

The content reflects pure psychological coercion, with no evidence of technical compromise.

Ensure you have sufficient funds to cover your payments

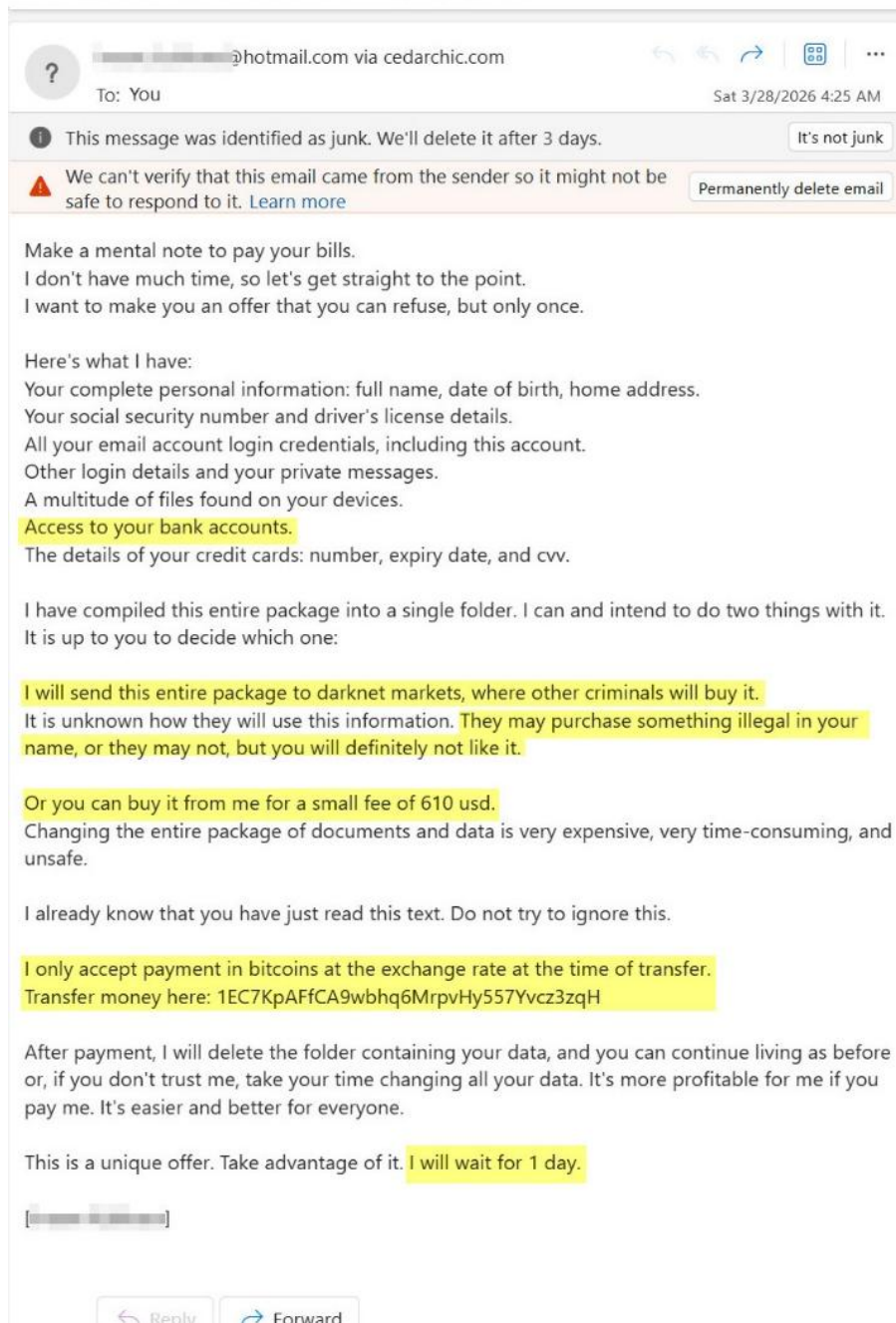


Figure 1: Sextortion email excerpt highlighting psychological coercion, claims of sensitive data compromise, financial demand, and Bitcoin-based payment instructions.

Additional examples of similar email content are provided in Appendix A.

4. Infrastructure & Tactics Analysis

4.1 Overview

The analysis of the five investigated email samples reveals the use of a distributed, heterogeneous, and infrastructure-aware operational model leveraging multiple cloud service providers, diverse domain configurations, and varied email delivery mechanisms. This architectural fragmentation is not incidental; rather, it reflects a deliberate and controlled strategy designed to evade detection, reduce attribution, and ensure operational continuity even in the event of partial infrastructure disruption or indicator exposure.

At the technical level, key email header components—including originating IP addresses, SMTP relay paths, HELO/EHLO identifiers, and authentication results—provide critical visibility into the underlying delivery infrastructure, as shown in Figure 2. These elements form the foundation for tracing message origin, identifying infrastructure characteristics, and assessing the legitimacy of the sending environment.

The originating IP addresses consistently resolve to cloud-hosted Virtual Private Server (VPS) environments distributed across multiple geographic regions and providers. This aligns with common adversarial practices, where attackers leverage low-cost, rapidly deployable infrastructure that can be easily rotated or abandoned with minimal operational impact. The diversity of providers and locations further complicates correlation efforts and reduces the effectiveness of geo-based or provider-based filtering.

Analysis of email authentication mechanisms reveals systematic inconsistencies across the samples. In multiple instances, SPF, DKIM, and DMARC validation results are either absent, misaligned, or explicitly fail verification checks. These failures indicate that the sending infrastructure is not authorized to transmit messages on behalf of the domains presented in the email headers or visible sender fields. This misalignment between claimed identity and actual sending source constitutes clear evidence of email spoofing and domain impersonation.

In addition to authentication failures, deeper inspection of SMTP “Received” header chains exposes irregular relay patterns. Several messages demonstrate direct transmission from VPS-hosted servers without passing through properly configured mail transfer agents. The relay paths often lack expected intermediate infrastructure, suggesting either simplified delivery chains or deliberate omission of standard routing practices.

Further anomalies are observed in HELO/EHLO identifiers, where generic or non-standard values such as “localhost” or default system hostnames are used instead of fully qualified domain names associated with legitimate mail servers. Such identifiers are characteristic of minimally configured or intentionally misconfigured SMTP services and are frequently associated with malicious or unauthorized email distribution.

The combination of these factors—unauthorized sending infrastructure, authentication failures, irregular relay paths, and anomalous SMTP identifiers—provides strong technical evidence of non-legitimate email delivery. More importantly, these characteristics are consistently observed across multiple samples despite variation in specific indicators, reinforcing the presence of a coordinated operational approach.

Collectively, the header analysis demonstrates that the campaign relies on flexible, low-attribution infrastructure combined with deliberately weak or invalid email authentication configurations. This approach enables scalable distribution while minimizing traceability and maximizing the likelihood of bypassing basic email security controls that rely on static validation mechanisms.

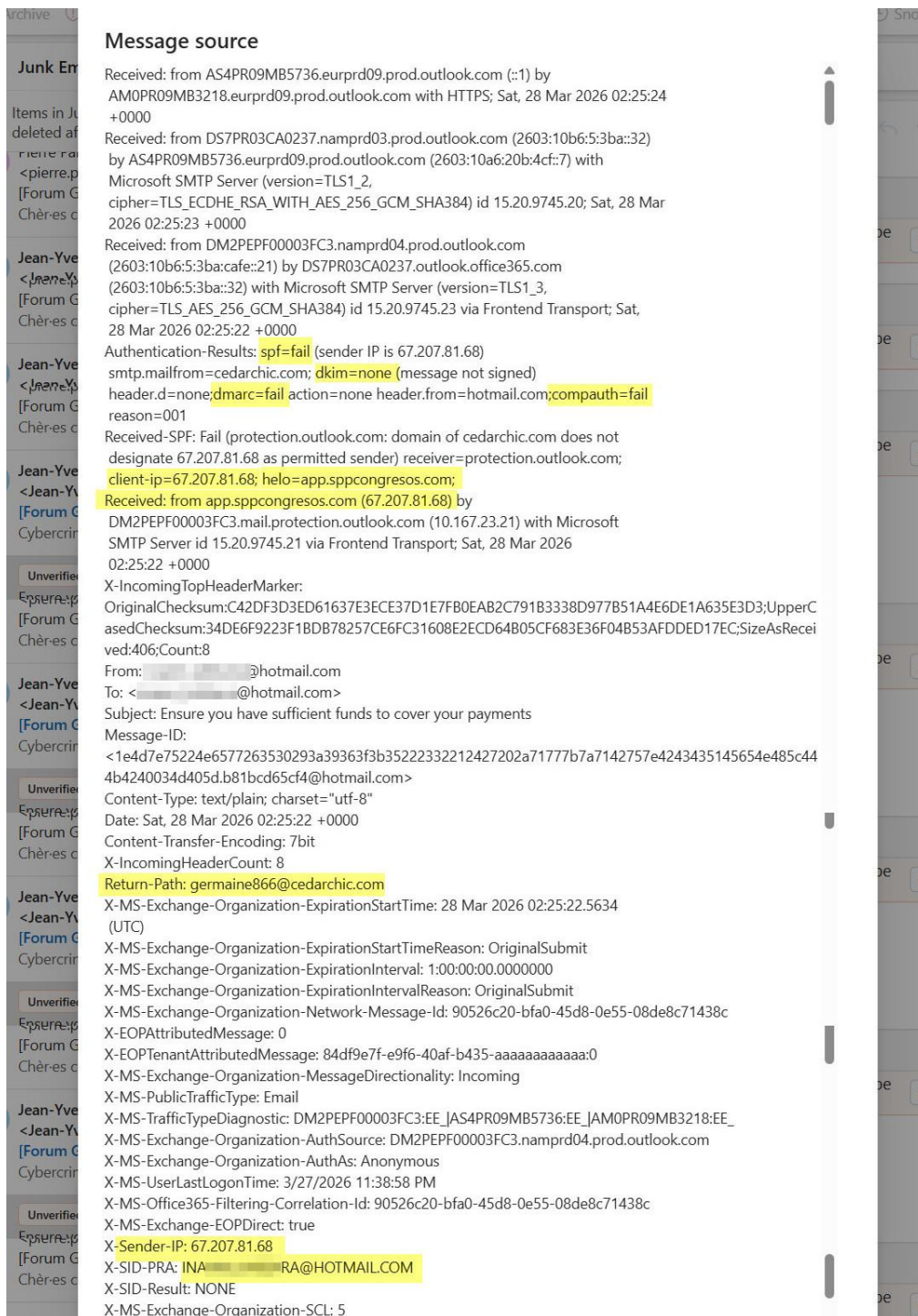


Figure 2: Extract of email header showing originating IP, SMTP path, and authentication results used for infrastructure analysis

Additional email header samples demonstrating variations in SMTP relay paths, originating infrastructure identification, and authentication outcomes across the analyzed emails are provided in Appendix B.

4.2 IP Infrastructure Analysis

All observed emails were transmitted from cloud-based hosting environments, including providers such as DigitalOcean and Linode, as well as a regional hosting provider located in Malaysia. The identified IP addresses correspond to Virtual Private Server (VPS) deployments within data center networks, spanning multiple geographic regions, including the United States, Singapore, and Malaysia, as shown in Figure 3.

IPinfo 67.207.81.68 Products Data Why IPinfo? Pricing Resources Docs Login Sign up

All IP Ranges > 67.0.0.0/8 > 67.207.0.0/16 > 67.207.81.0/24 > 67.207.81.68

67.207.81.68 Star

North Bergen, New Jersey, US Cloud Hosting SSH Webserver

Need more data or want to access it via API or data downloads? Sign up to get free access Sign up for free

Summary

ASN	AS14061 - DigitalOcean, LLC
Hostname	No Hostname
Range	67.207.80.0/20
Company	DigitalOcean, LLC
Hosted domains	0
Privacy	True
Anycast	False
ASN type	Hosting
Abuse contact	abuse@digitalocean.com

IP Geolocation

City	North Bergen
State	New Jersey
Country	United States
Postal	07047
Local time	02:09 PM, Saturday, May 02, 2026
Timezone	America/New_York
Coordinates	40.8043 N, 74.0121 W

Map data ©2026 Google Terms Report a map error

40.8043 N, 74.0121 W

IP Geolocation data

IP geolocation lookup is the identification of an IP address' geographic location in the real world.

Useful for Web Personalization, and Financial Technology

IP Geolocation API & Database >

Figure 3: IP intelligence lookup showing cloud-hosted infrastructure, ASN association, and geographic location of the originating mail server.

Additional IP intelligence results across the analyzed samples are provided in Appendix B (Figures B1–B4).

The consistent reliance on VPS infrastructure indicates the use of rented, easily replaceable resources, a well-established characteristic of malicious campaigns due to

their low cost, scalability, and limited attribution. This infrastructure model enables rapid provisioning and decommissioning of sending nodes, thereby reducing the operational impact of detection, blocking, or blacklisting.

From an attribution perspective, the use of geographically distributed cloud infrastructure introduces an additional layer of obfuscation, as the physical location of the server does not reliably reflect the actual location of the threat actor. Instead, it reflects infrastructure availability and deployment strategy.

Reputation analysis conducted via AbuseIPDB reveals a mixed profile across the identified IP addresses. Some addresses show recent reports associated with phishing, spam, and extortion activity, confirming active malicious usage. In contrast, other IPs exhibit low or no prior reporting, suggesting newly deployed or recently rotated infrastructure, as shown in Figure 4.

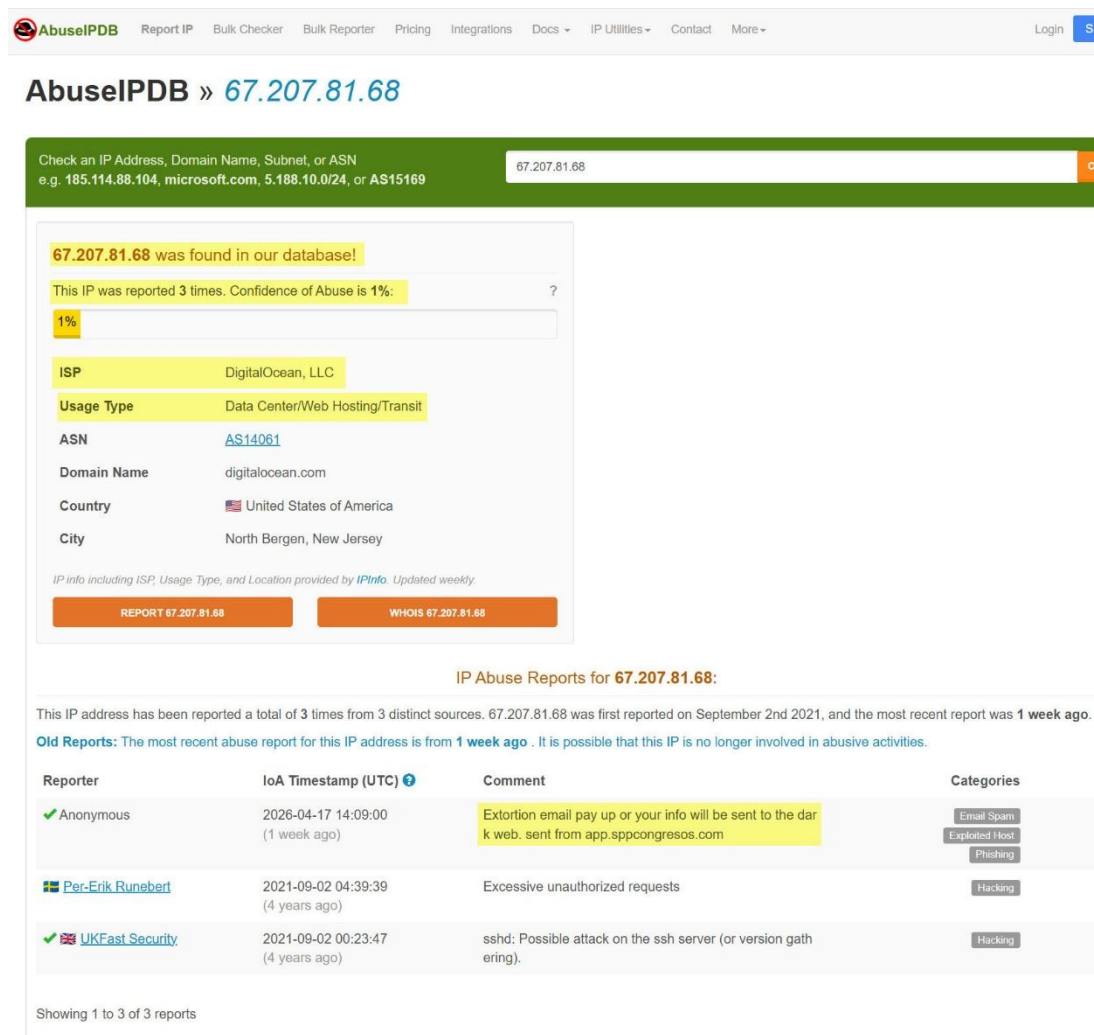


Figure 4: AbuseIPDB results showing historical abuse reports and classification of extortion and phishing activity associated with the analyzed IP address.

Additional reputation analysis results are provided in Appendix B (Figures B5–B8).

This variation strongly indicates a blended operational strategy in which previously flagged infrastructure is reused alongside newly provisioned or low-reputation assets. Such a hybrid approach reduces detection rates, delays blacklisting, and enhances the overall persistence of the campaign.

4.3 Domain and Infrastructure Configuration

The analysis identified multiple domain usage patterns, reflecting a flexible and adaptive approach to infrastructure deployment across the campaign.

Certain domains exhibit characteristics consistent with disposable or low-attribution assets, including privacy-protected registration, limited ownership visibility, and hosting environments aligned with cloud-based providers. These domains are likely acquired specifically for malicious use and can be abandoned with minimal operational cost, as shown in Figures 5 and 6.

 Domains Hosting Websites Email Security Whois Deals

cedarchic.com

Updated 18 seconds ago

 Domain Information

Domain:	cedarchic.com
Registered On:	2023-10-31
Expires On:	2026-10-31
Updated On:	2025-10-25
Status:	client transfer prohibited
Name Servers:	ns1.afternic.com ns2.afternic.com verification-qxu9bfdaih4cxurbbn3f4d.ns101.verify.hn

 Registrar Information

Registrar:	Dynadot Inc
IANA ID:	472
Email:	abuse@dynadot.com
Abuse Email:	abuse@dynadot.com
Abuse Phone:	+16502620100

 Registrant Contact

Name:	Dynadot Privacy Service
Organization:	Super Privacy Service LTD c/o Dynadot
Street:	PO Box 701
City:	San Mateo
State:	California
Postal Code:	94401
Country:	US
Phone:	+1.6505854708
Email:	https://www.dynadot.com/domain/contact-request?domain=cedarchic.com

Figure 5: WHOIS domain lookup (return-path) showing privacy-protected registration, registrar details, and anonymized ownership information, indicating potential identity obfuscation.

sppcongresos.com

Updated 1 second ago  Error encountered. Please try again later.

Domain Information

Domain:	sppcongresos.com
Registered On:	2022-05-29
Expires On:	2026-05-29
Updated On:	2025-06-03
Status:	client transfer prohibited
Name Servers:	ns1.digitalocean.com ns2.digitalocean.com ns3.digitalocean.com



Registrar Information

Registrar:	NameCheap, Inc.
IANA ID:	1068
Email:	support@namecheap.com
Abuse Email:	abuse@namecheap.com
Abuse Phone:	+1.6613102107



Registrant Contact

Organization:	Privacy service provided by Withheld for Privacy ehf
Street:	Kalkofnsvegur 2
City:	Reykjavik
State:	Capital Region
Postal Code:	101
Country:	IS
Phone:	+354.4212434
Email:	debb39ff9dcb455c852210d38af92902.protect@withheldforprivacy.com

Figure 6: WHOIS domain lookup showing privacy-protected registration, registrar details, and anonymized ownership information, indicating potential identity obfuscation.

Additional WHOIS records and domain registration data across the analyzed samples are provided in Appendix C (Figures C1–C5).

In other cases, emails leverage subdomains associated with older or established domains. This pattern suggests either compromise of legitimate domains or unauthorized use of delegated subdomain configurations. Such techniques enable attackers to embed malicious activity within otherwise legitimate domain structures, increasing the likelihood of bypassing basic filtering mechanisms that rely on domain reputation.

Additional obfuscation is achieved through the use of proxy and reverse proxy services, such as Cloudflare, which conceals the underlying hosting infrastructure. By routing traffic through intermediary layers, attackers introduce additional complexity into attribution efforts and reduce the visibility of the true origin servers, thereby limiting direct infrastructure correlation.

The campaign also demonstrates clear instances of domain spoofing and impersonation. Certain emails reference legitimate domains that show no technical linkage to the sending infrastructure, confirming that these identities are fabricated and not under attacker control. This misalignment between visible sender identity and actual infrastructure further reinforces the use of deception as a core operational tactic.

Furthermore, at least one sample utilizes a non-existent or unregistered return-path domain, indicating a complete absence of legitimate email configuration. This reflects a low-level SMTP usage model in which valid domain ownership, proper DNS configuration, and authentication alignment are not required, further highlighting the adversary's reliance on permissive or weakly enforced email validation mechanisms.

4.4 Email Delivery Techniques

Beyond domain variation, the campaign employs multiple email delivery techniques. Emails are sent through standard SMTP services hosted on VPS environments, as well as through subdomain-based configurations and proxy-routed infrastructures.

Notably, some samples reveal direct SMTP transmission using “localhost” HELO identifiers and default hosting provider hostnames instead of properly configured mail server identities. This indicates the use of minimally configured or deliberately misconfigured SMTP services.

The presence of such generic identifiers reflects low-level email delivery operations that bypass standard mail server conventions, thereby reducing traceability and complicating attribution efforts.

4.5 Infrastructure Correlation

Despite the observed variation in domains, IP addresses, and delivery techniques, a consistent operational pattern emerges. Cloud hosting providers are repeatedly utilized across samples, and domain configurations sometimes align with the underlying infrastructure while in other cases remain completely decoupled, particularly when proxy services or spoofed identities are involved.

This variability does not indicate inconsistency but rather demonstrates a modular deployment model in which infrastructure components can be selectively combined, replaced, or discarded as needed. Such flexibility enhances the resilience of the campaign and allows it to adapt dynamically to defensive measures.

4.6 Key Observations

The campaign is characterized by a multi-provider infrastructure spanning multiple geographic regions, combined with rotating IP usage and mixed reputation visibility. Domain strategies are diverse, including disposable domains, compromised or abused subdomains, spoofed identities, and non-existent configurations. These are complemented by multiple email delivery techniques, including direct SMTP abuse.

The intentional inconsistency across these elements reflects a deliberate effort to evade detection, reduce correlation between campaign components, and maintain operational persistence.

4.7 Analytical Conclusion

The observed activity reflects a moderately advanced, infrastructure-aware email campaign that leverages distributed cloud resources, fragmented identity mechanisms, and adaptive delivery techniques to achieve scalability, resilience, and evasion.

The inconsistencies observed across IP reputation, domain registration, and email configuration should not be interpreted as operational weaknesses. On the contrary, they represent deliberate design choices aimed at complicating detection and attribution, while enabling sustained and scalable malicious activity.

5. Blockchain Transaction Analysis

5.1 Objective

The objective of this section is to analyze the cryptocurrency wallets provided within the investigated emails; trace fund flows across multiple transactional hops and identify behavioral patterns indicative of a coordinated financial infrastructure.

5.2 Overview of Observed Activity

Analysis of the Bitcoin wallets associated with the emails reveals a consistent and structured pattern of activity. Transactions are characterized by multi-input aggregation, rapid forwarding of funds across successive wallets, and the absence of balance retention across all analyzed addresses. Additionally, intermediary wallets are reused across multiple emails, indicating operational linkage.

Across all samples, the wallets function as **transit nodes rather than storage endpoints**, suggesting that their primary purpose is to facilitate the movement and consolidation of funds rather than long-term holding.

5.3 Per-Email Wallet Chain Analysis

Email 1 – Wallet Chain

Wallet 1 (1EC7KpAFfCA9wbhq6MrpvHy557Yvcz3zqH)

Role: Initial collection wallet

This wallet received and transferred an equal total amount of 0.05529138 BTC, indicating no balance retention. It participated in a 7-input aggregation transaction, combining funds from multiple sources before forwarding them to a subsequent wallet. (as shown in figure 7)

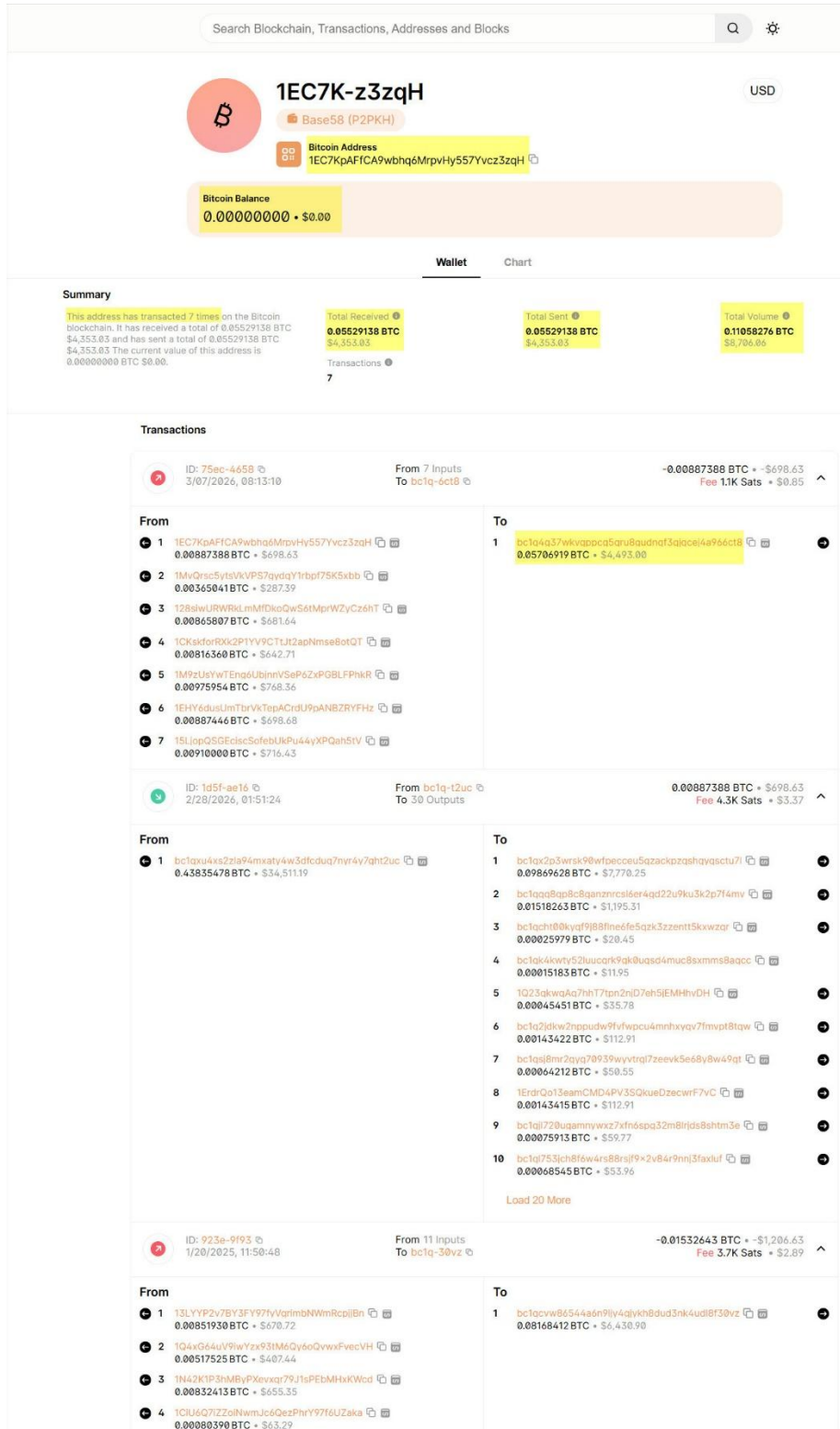


Figure 7: Blockchain view of Bitcoin wallet 1EC7KpAFfCA9wbhq6MrpvHy557Yvcz3zqH identified in email 1, showing incoming and outgoing transactions associated with initial fund collection

Wallet 2 (bc1q4g37wkvgppcg5qru8gudnqf3qjqcej4a966ct8)

Role: Aggregation / relay wallet

This wallet received consolidated funds totaling 0.05706919 BTC. It combined inputs from multiple upstream wallets and forwarded the entire balance in a single transaction, again with no retained funds. (as shown in figure 8)

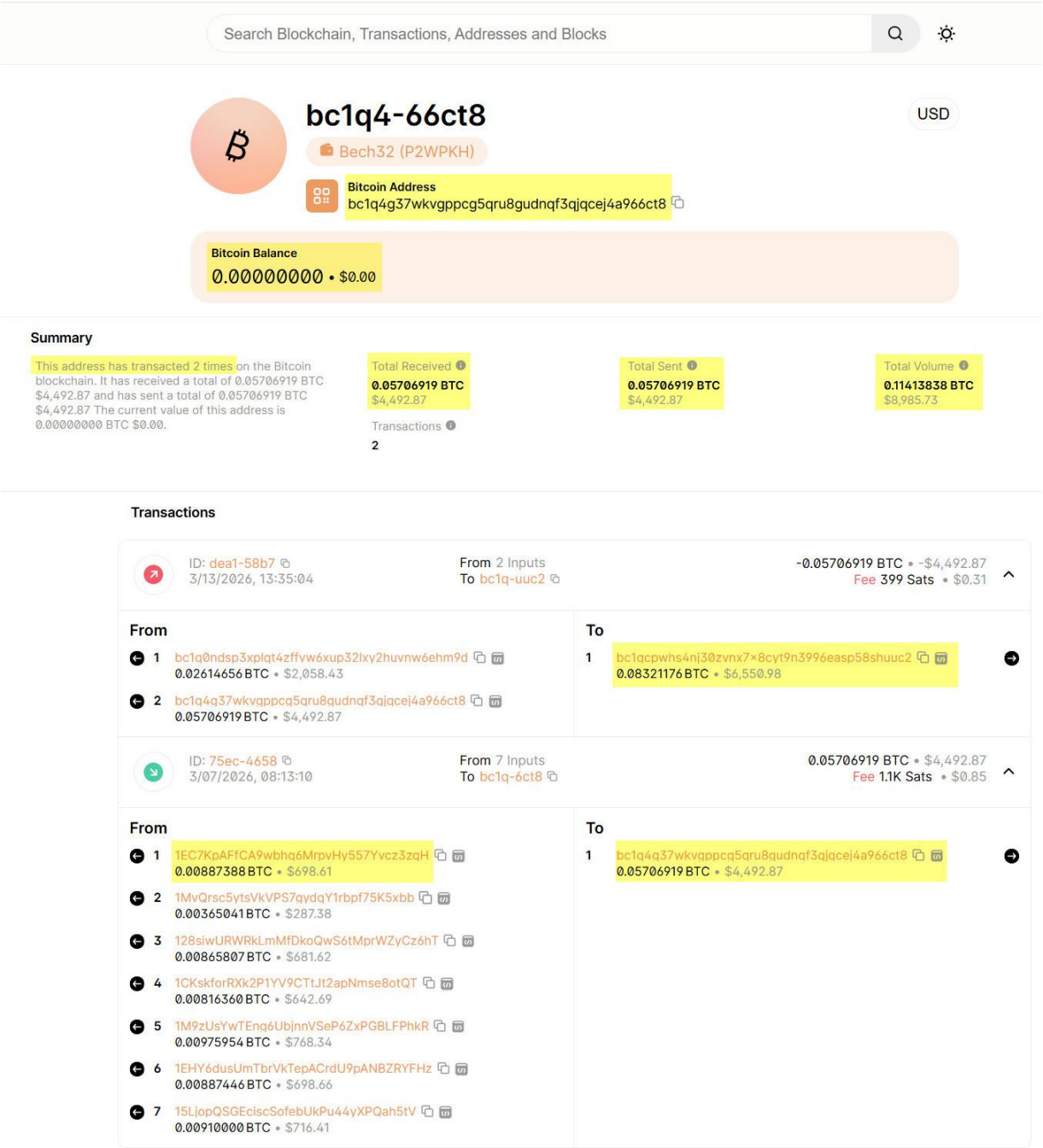


Figure 8: Transaction activity of Bitcoin wallet bc1q4g37wkvgppcg5qru8gudnqf3qjqcej4a966ct8 demonstrating aggregation and full transfer of received funds

Wallet 3 (bc1qcpwhs4nj30zvn7x8cyt9n3996easp58shuuc2)

Role: Consolidation / distribution wallet

This wallet received 0.08321176 BTC through aggregated inputs and immediately forwarded the funds to another address. No balance retention was observed. (as shown in figure 9)

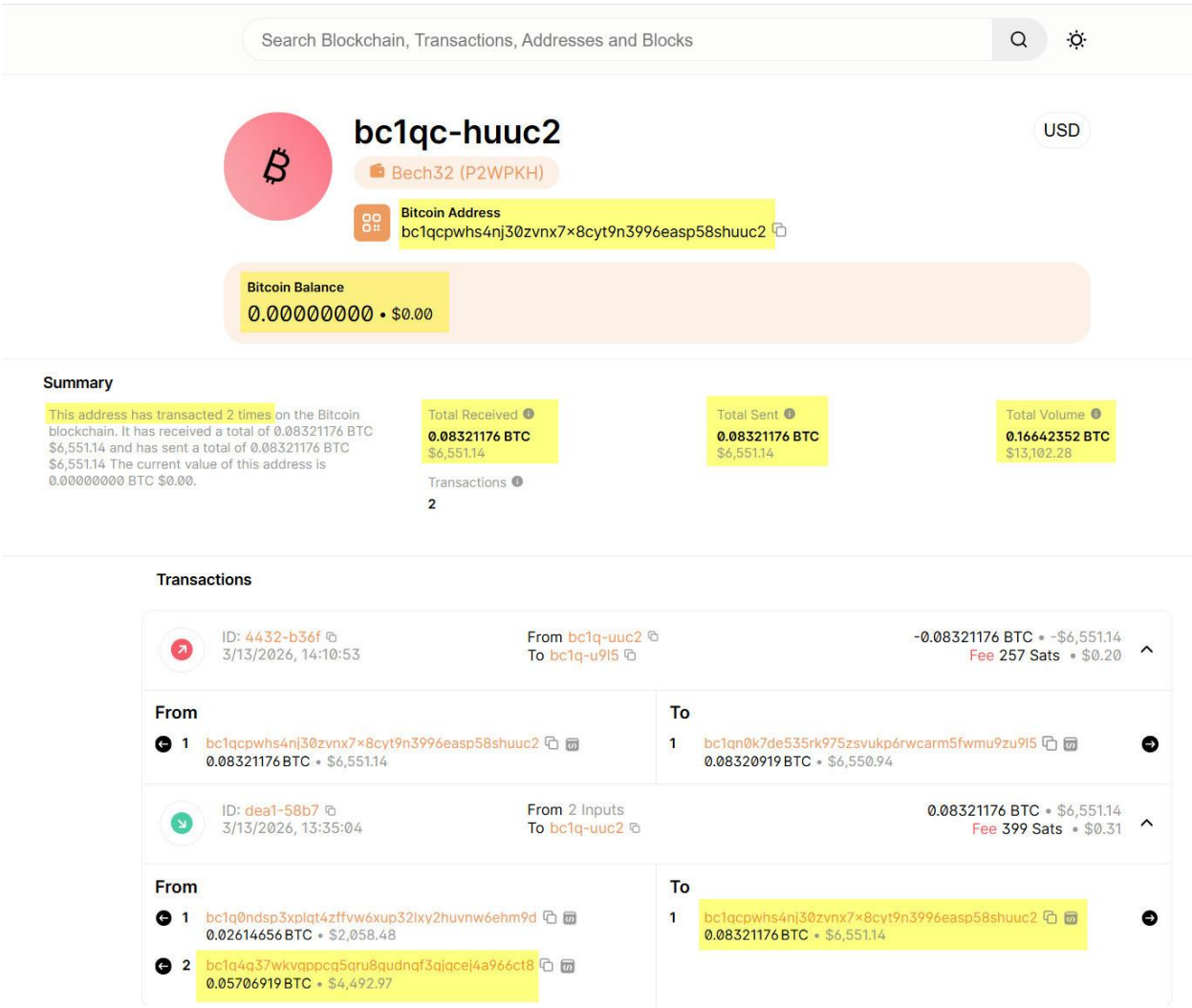


Figure 9: Blockchain records of Bitcoin wallet bc1qcpwhs4nj30zvn7x8cyt9n3996easp58shuuc2 illustrating consolidated fund intake and immediate redistribution

Email 2 – Wallet Chain

Wallet 1 (1CWTFeMfPCG1Q6uVLSpHUmQ1J1i6hxj1LK)

Role: Initial collection wallet

This wallet participated in a 9-input aggregation transaction and forwarded funds to a

Bech32 relay wallet. Its behavior closely mirrors that of the initial wallet observed in email 1. (as shown in figure 10)

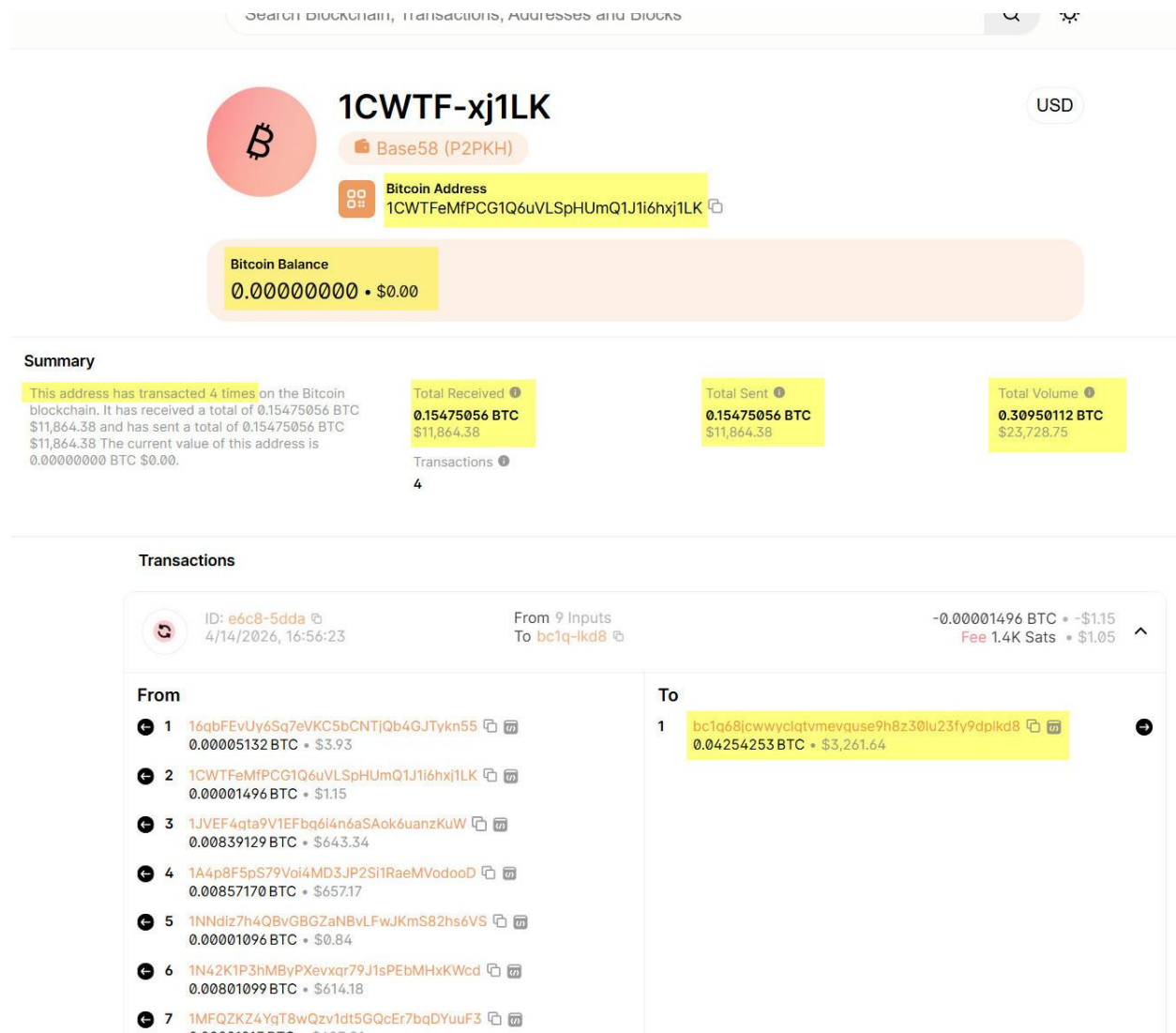


Figure 10: Bitcoin wallet 1CWTFeMfPCG1Q6uVLSpHUmQ1J1i6hxj1LK identified in email 2, showing multi-input aggregation transaction behavior

Wallet 2 (bc1q68jcwvwyqlatvmevguse9h8z30lu23fy9dplkd8)

Role: Relay wallet

This wallet received 0.04254253 BTC and forwarded the entire amount in a single transaction, retaining no balance. (as shown in figure 11)

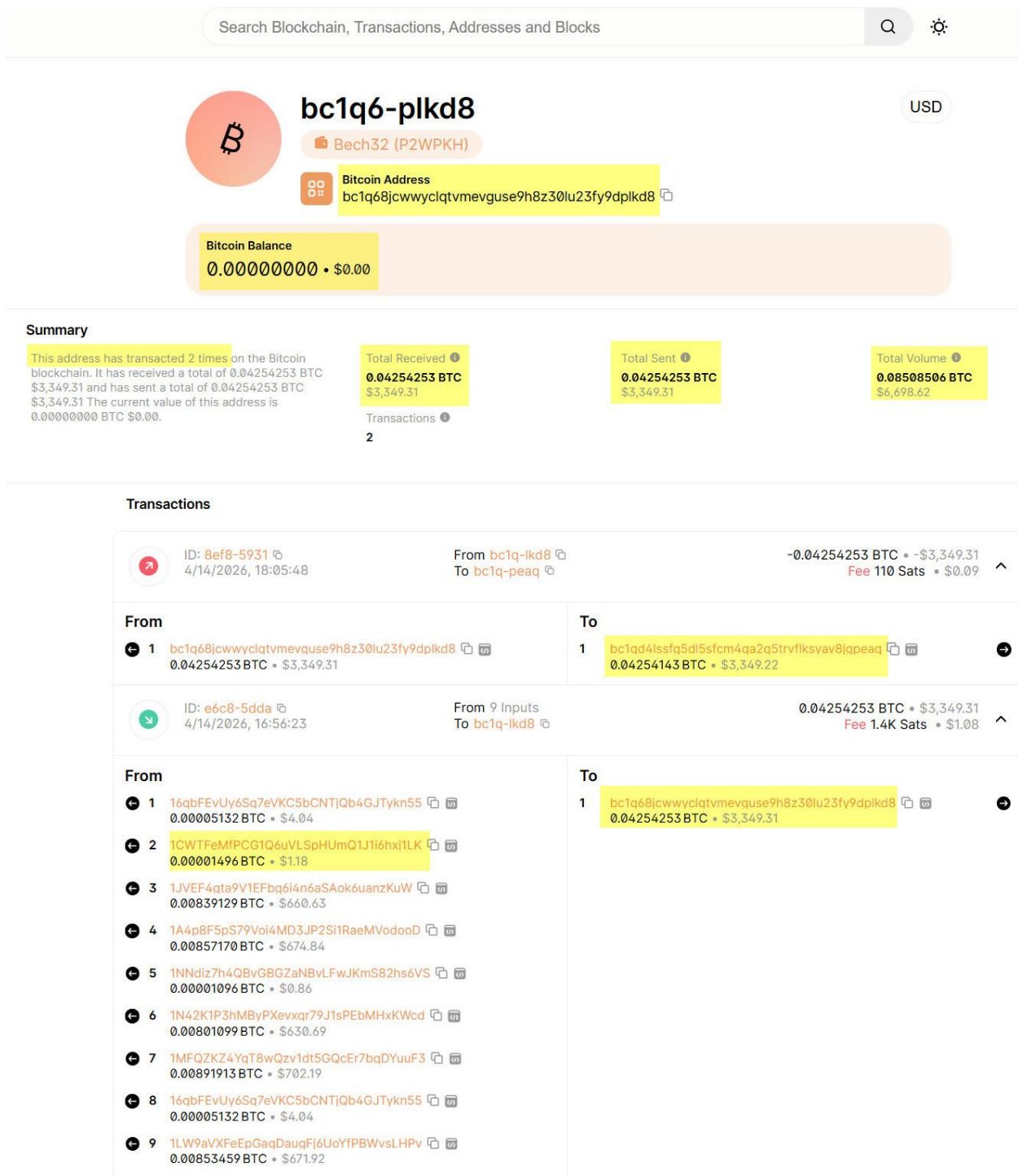


Figure 11: Transaction flow of Bitcoin wallet bc1q68jcwyclqtvmevguse9h8z30lu23fy9dplkd8 reflecting relay behavior with full balance forwarding

Wallet 3 (bc1qd4lssfg5dl5sfc4qa2q5trvflksyav8jgpeaq)

Role: Redistribution wallet

This wallet executed multi-output transactions, including smaller-value transfers alongside primary fund movements. This behavior suggests controlled redistribution and potential obfuscation of transaction flows. (as shown in figure 12)

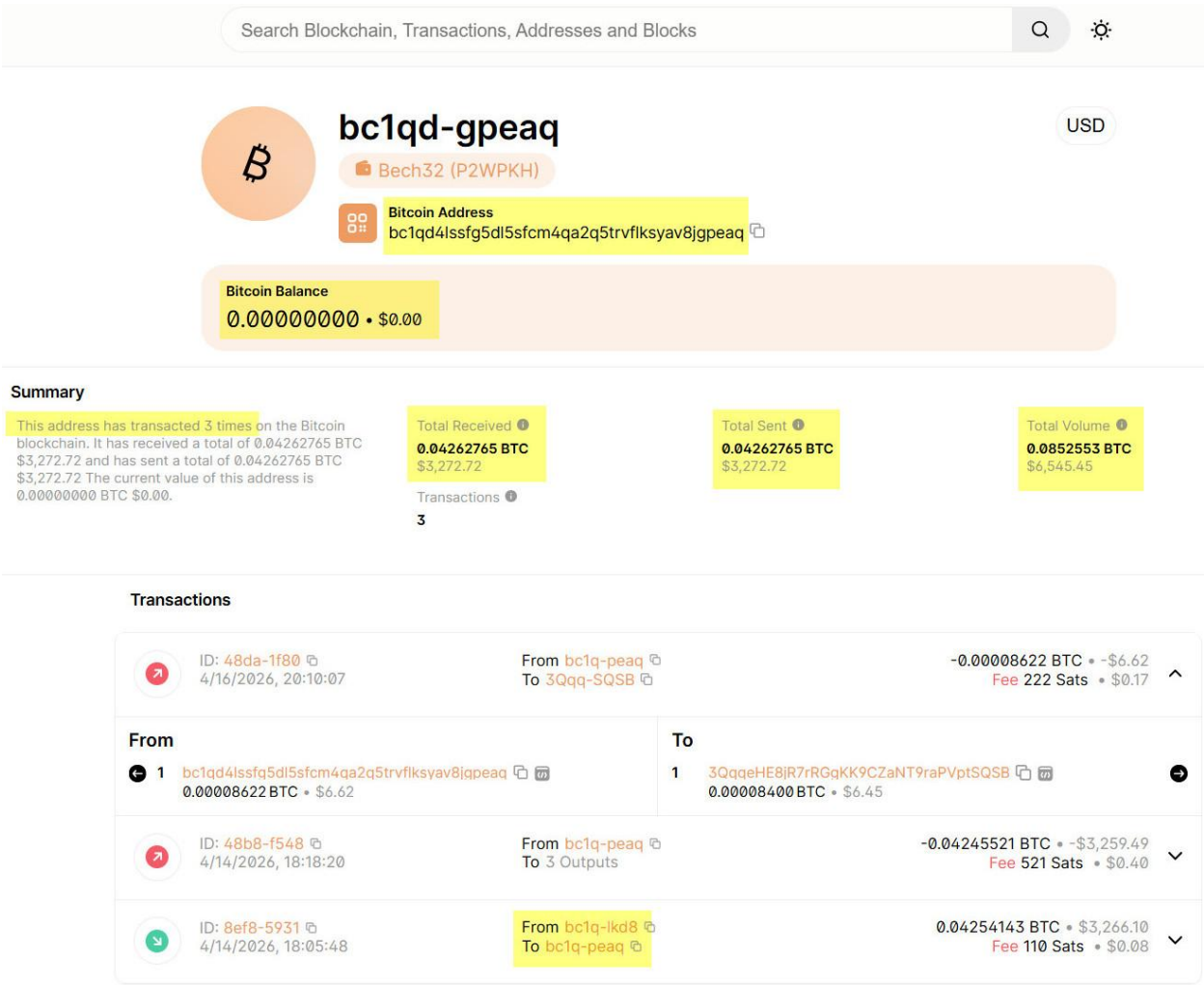


Figure 12: Multi-output transaction structure observed in Bitcoin wallet bc1qd4lssfg5dl5sfc4qa2q5trvflksyav8jgpeaq indicating redistribution and potential obfuscation patterns

Email 3 – Wallet Chain

Wallet 1 (19tWNhBbjDHBHk9FqQzhU5k7w2E9pRQrrU)

Role: Initial collection wallet

This wallet participated in a 9-input aggregation transaction, incorporating inputs from multiple external sources. Notably, one transaction includes input from a labeled

exchange source (“Binance 4”), indicating interaction with external financial platforms. (as shown in figure 13)

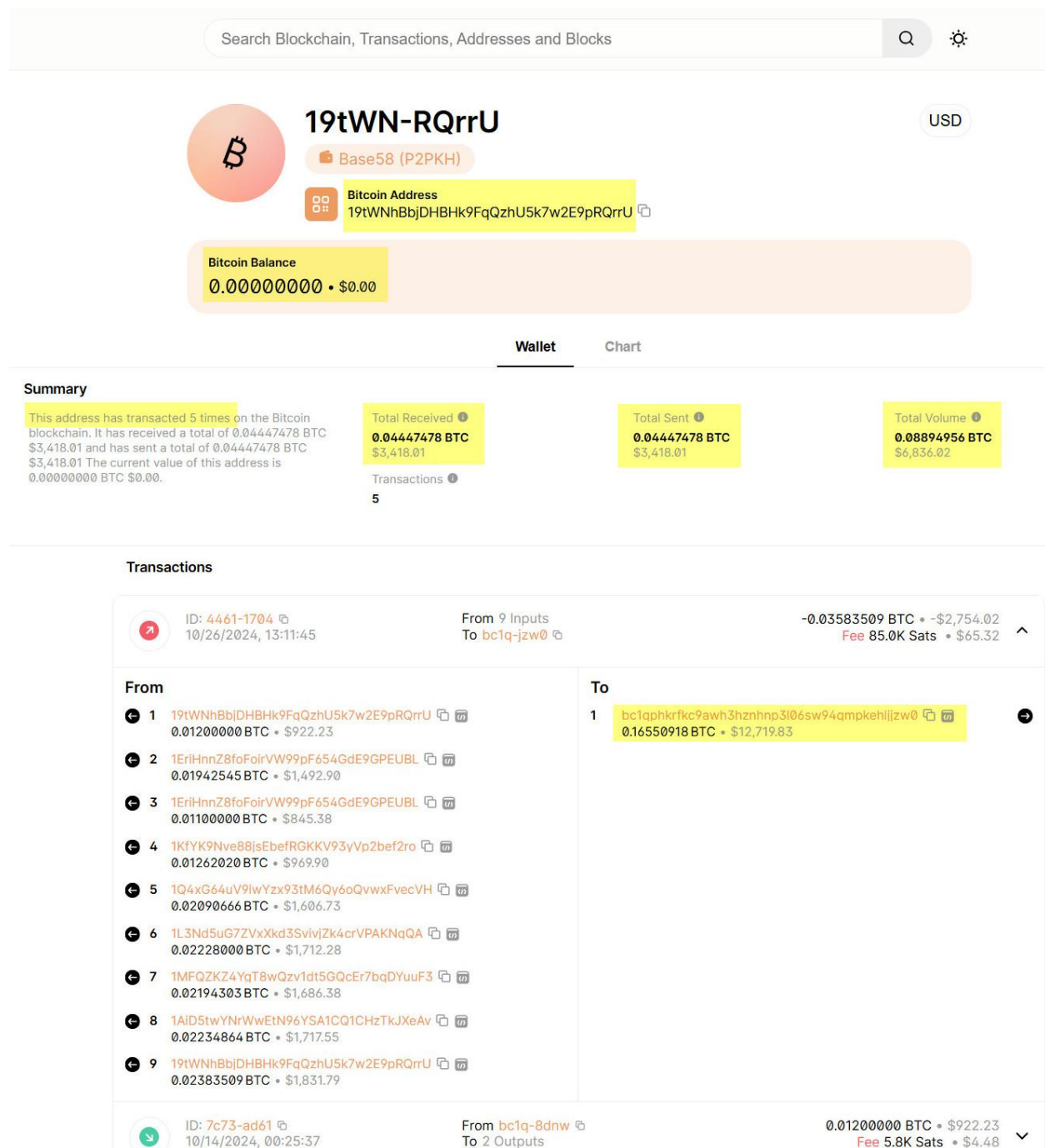


Figure 13: Bitcoin wallet 19tWNhBbjDHBHk9FqQzhU5k7w2E9pRQrrU showing aggregated inputs including interaction with external exchange sources

Wallet 2 (bc1qphkrfk9awh3hznhnp3l06sw94qmpkehljjzw0)

Role: Central aggregation wallet

This wallet received a significant aggregated amount of 0.18109869 BTC and performed multiple outgoing transactions. It functions as a key node within the transaction network, consolidating funds before redistribution. (as shown in figure 14)

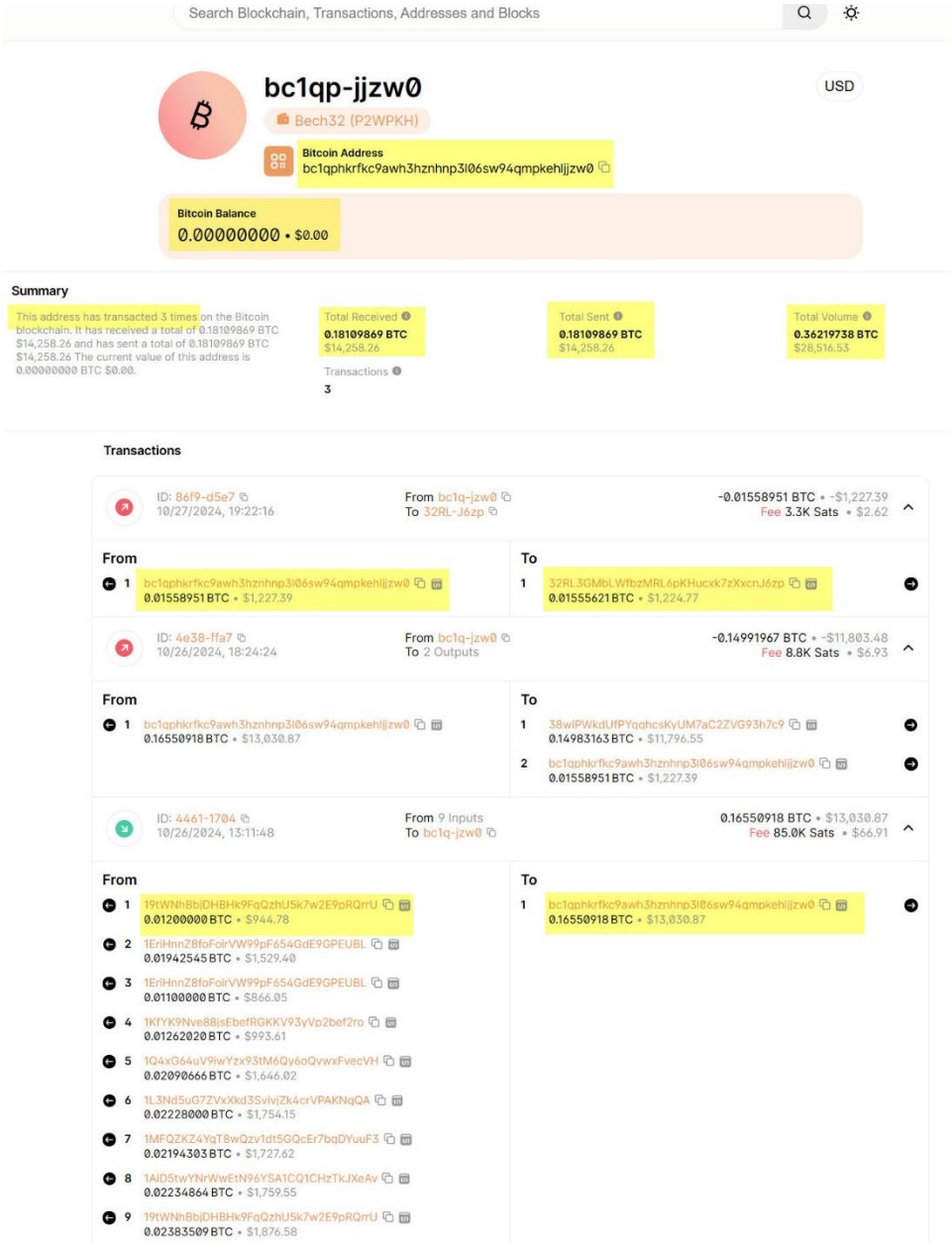


Figure 14: Central aggregation activity in Bitcoin wallet bc1qphkrfk9awh3hznhnp3l06sw94qmpkehljjzw0 consolidating significant transaction volumes

Wallet 3 (32RL3GMbLWfbzMRL6pKHucxk7zXxcnJ6zp)

Role: Downstream relay wallet

This wallet received funds from the central aggregation node and forwarded them immediately, maintaining zero balance. (as shown in figure 15)

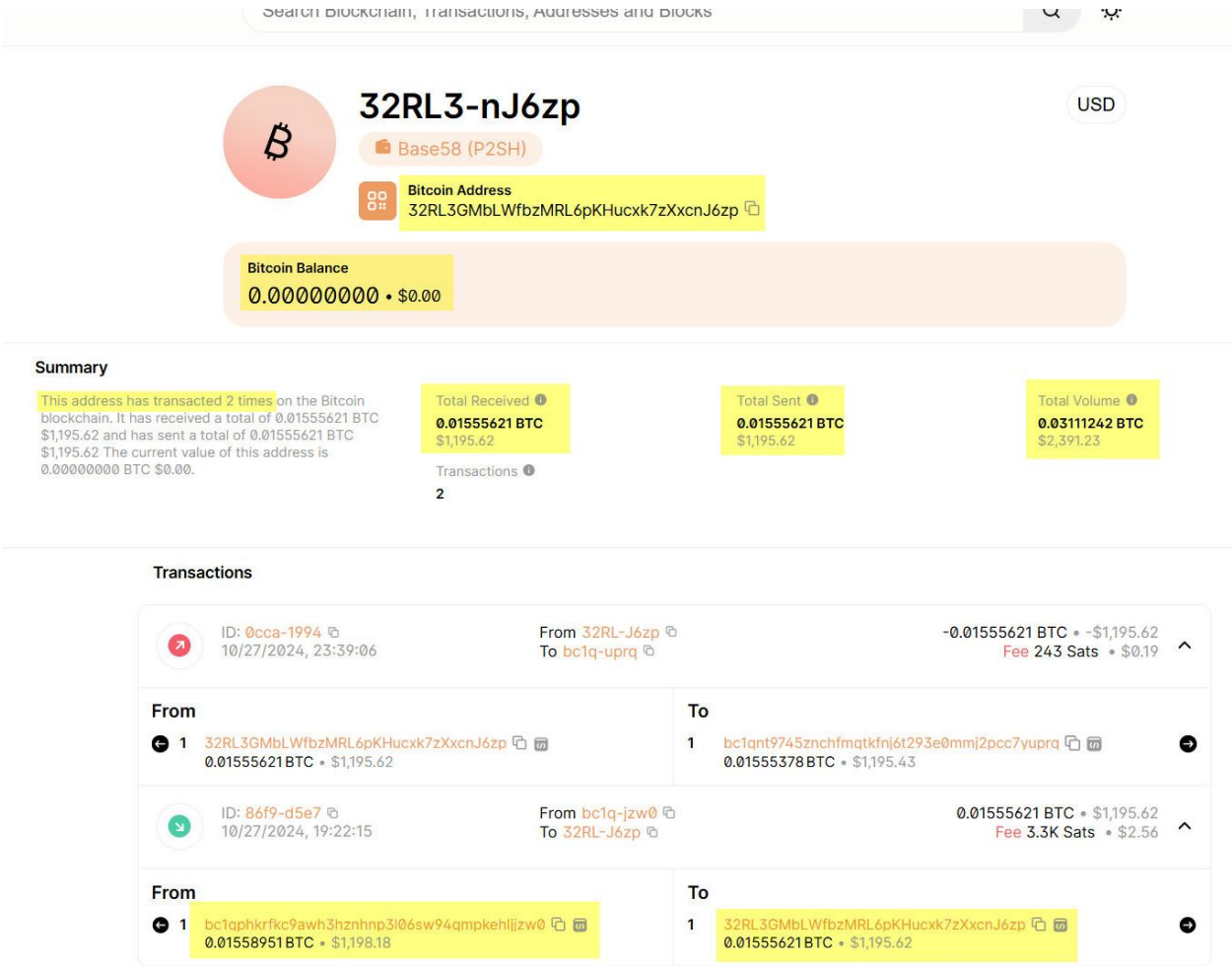


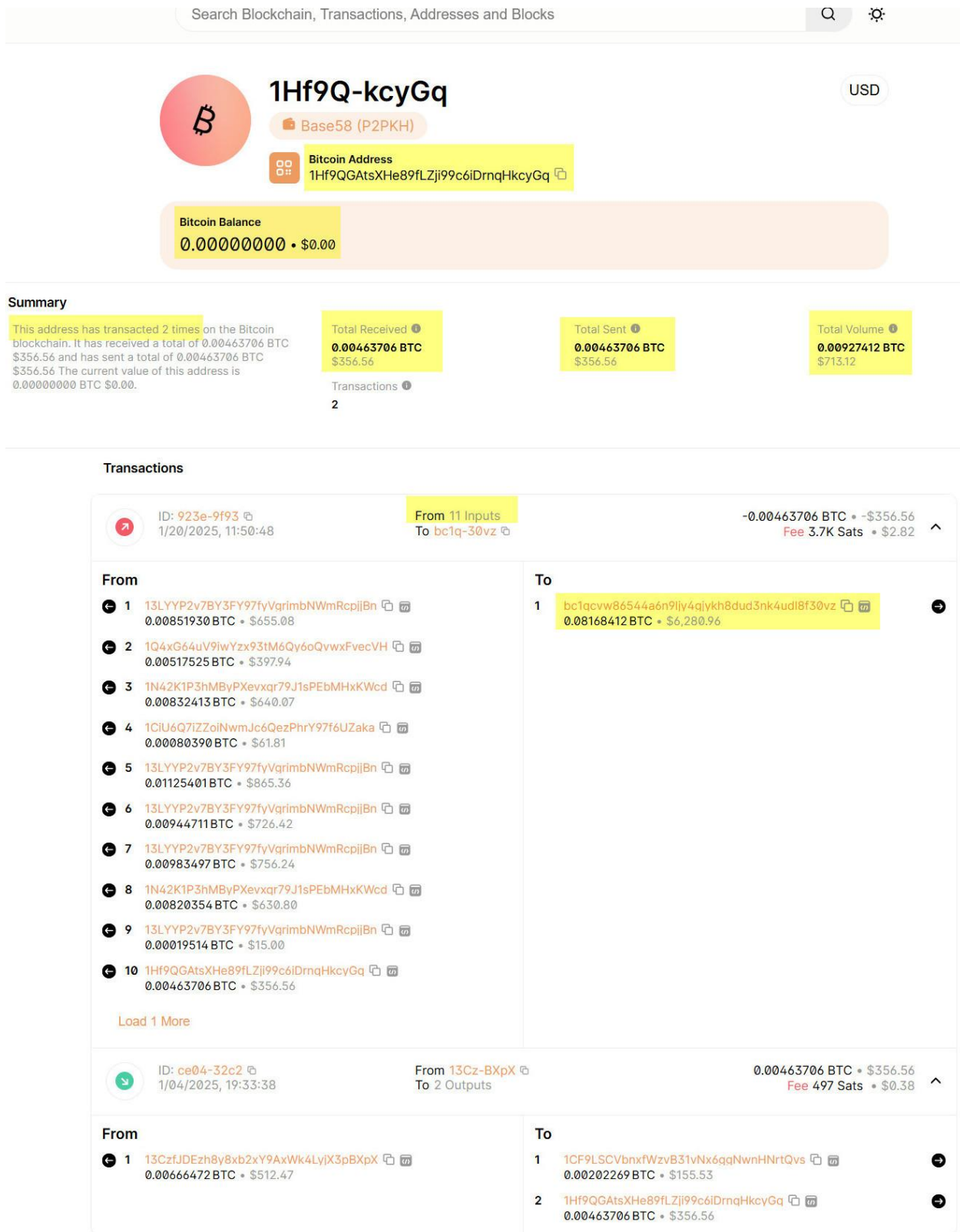
Figure 15: Downstream transaction flow from Bitcoin wallet 32RL3GMbLWfbzMRL6pKHucxk7zXxcnJ6zp demonstrating immediate forwarding behavior

Email 4 – Wallet Chain

Wallet 1 (1Hf9QGAtsXHe89fLZji99c6iDrnqHkcyGq)

Role: Initial collection wallet

This wallet participated in an 11-input aggregation transaction, contributing to a larger pooled transaction and forwarding funds without retention. (as shown in figure 16)



Wallet 2 (bc1qcvw86544a6n9ljy4gjykh8dud3nk4udl8f30vz)

Role: Aggregation wallet

This wallet received 0.08168412 BTC and forwarded the entire balance in a single transaction. (as shown in figure 17)

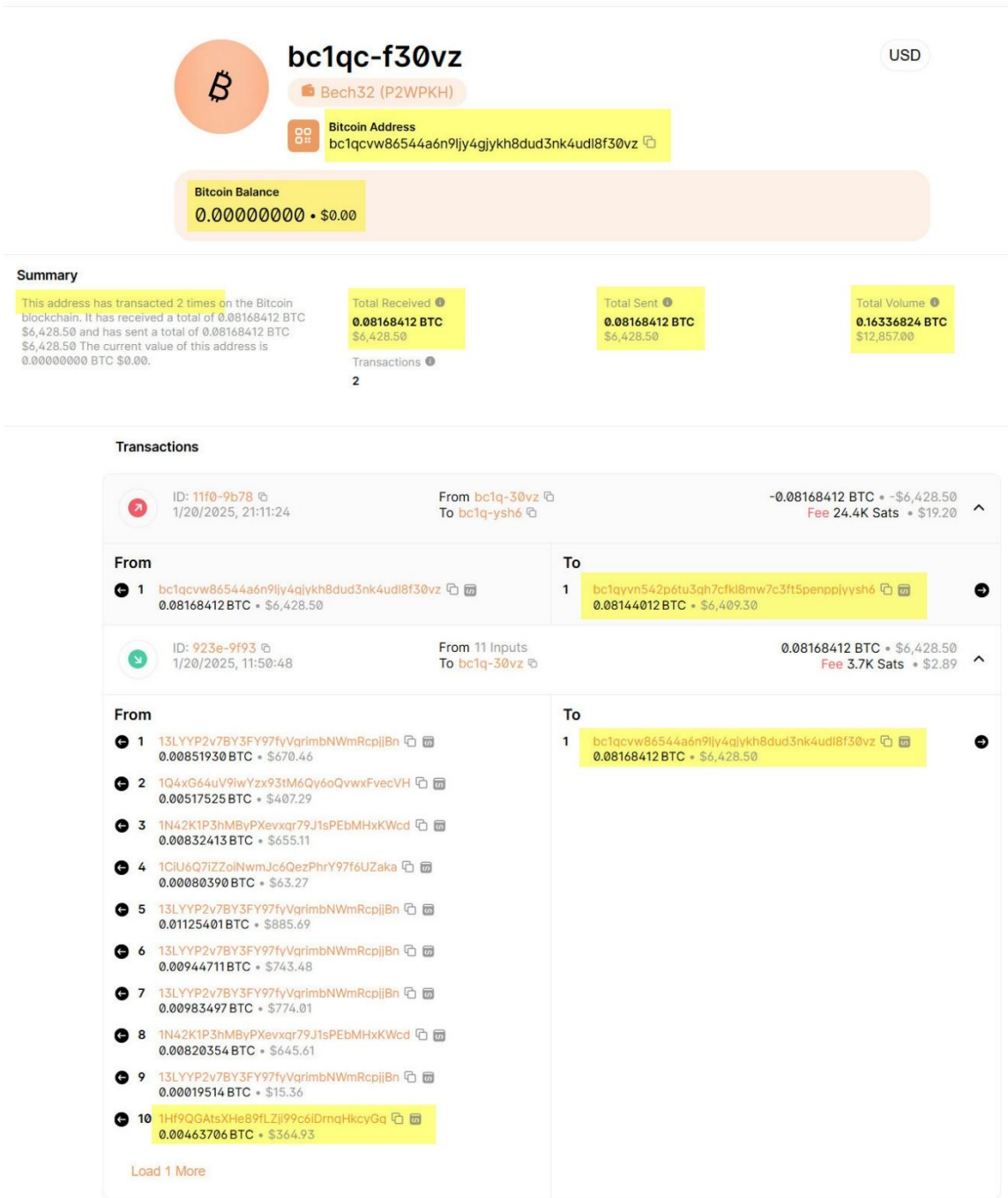


Figure 17: Aggregated fund transfer observed in Bitcoin wallet bc1qcvw86544a6n9ljy4gjykh8dud3nk4udl8f30vz showing complete forwarding of received balance

Wallet 3 (bc1qyv542p6tu3gh7cfkl8mw7c3ft5penppjy6)

Role: Relay wallet

This wallet continued the forwarding chain, with rapid transaction sequences and no balance retention observed. (as shown in figure 18)

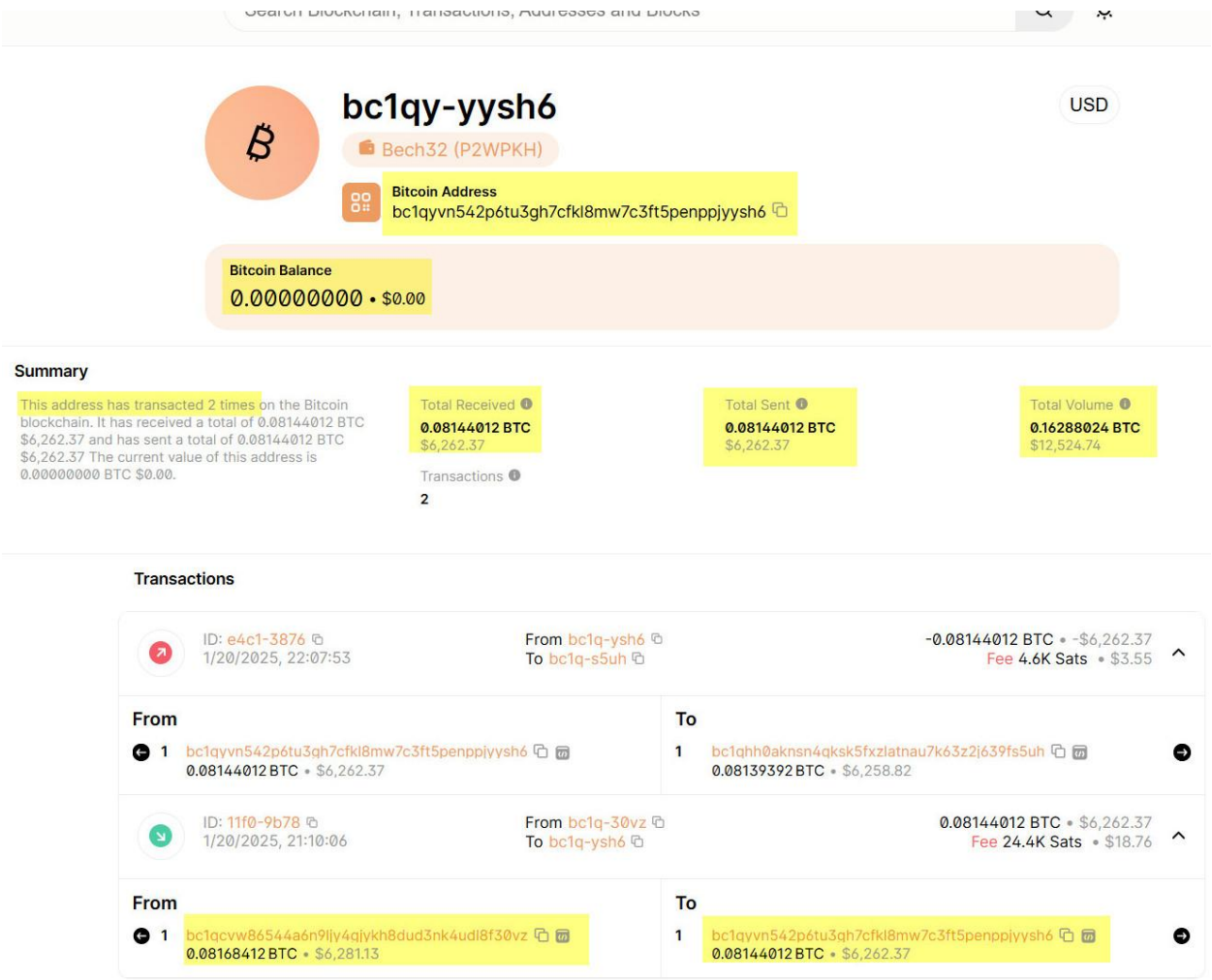


Figure 18: Relay transaction pattern in Bitcoin wallet bc1qyv542p6tu3gh7cfkl8mw7c3ft5penppjy6 with no retained funds

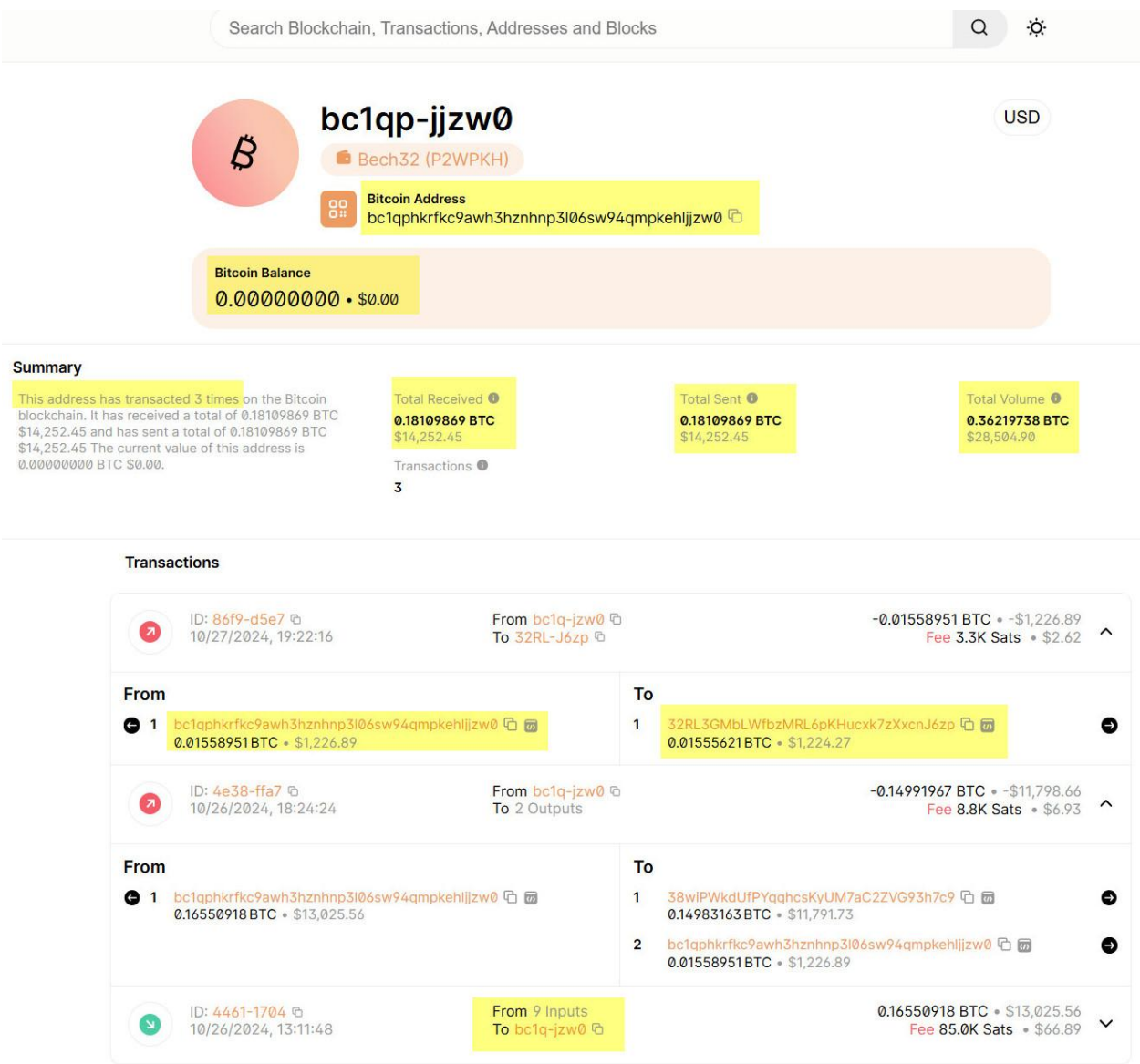
Email 5 – Wallet Chain

Wallet 1 (1L3Nd5uG7ZVxXkd3SvivjZk4crVPAKNqQA)

Role: Initial collection wallet

This wallet participated in a 9-input aggregation transaction, with one input originating from a labeled exchange source (“Binance 4”). It forwarded funds to a shared aggregation wallet. (as shown in figure 19)

Wallet 2 (bc1qphkrfk9awh3hznnp3l06sw94qmpkehljjzw0)
Role: Central aggregation wallet (reused)
This wallet is identical to the aggregation node identified in email 3, confirming reuse across multiple emails. (as shown in figure 20)



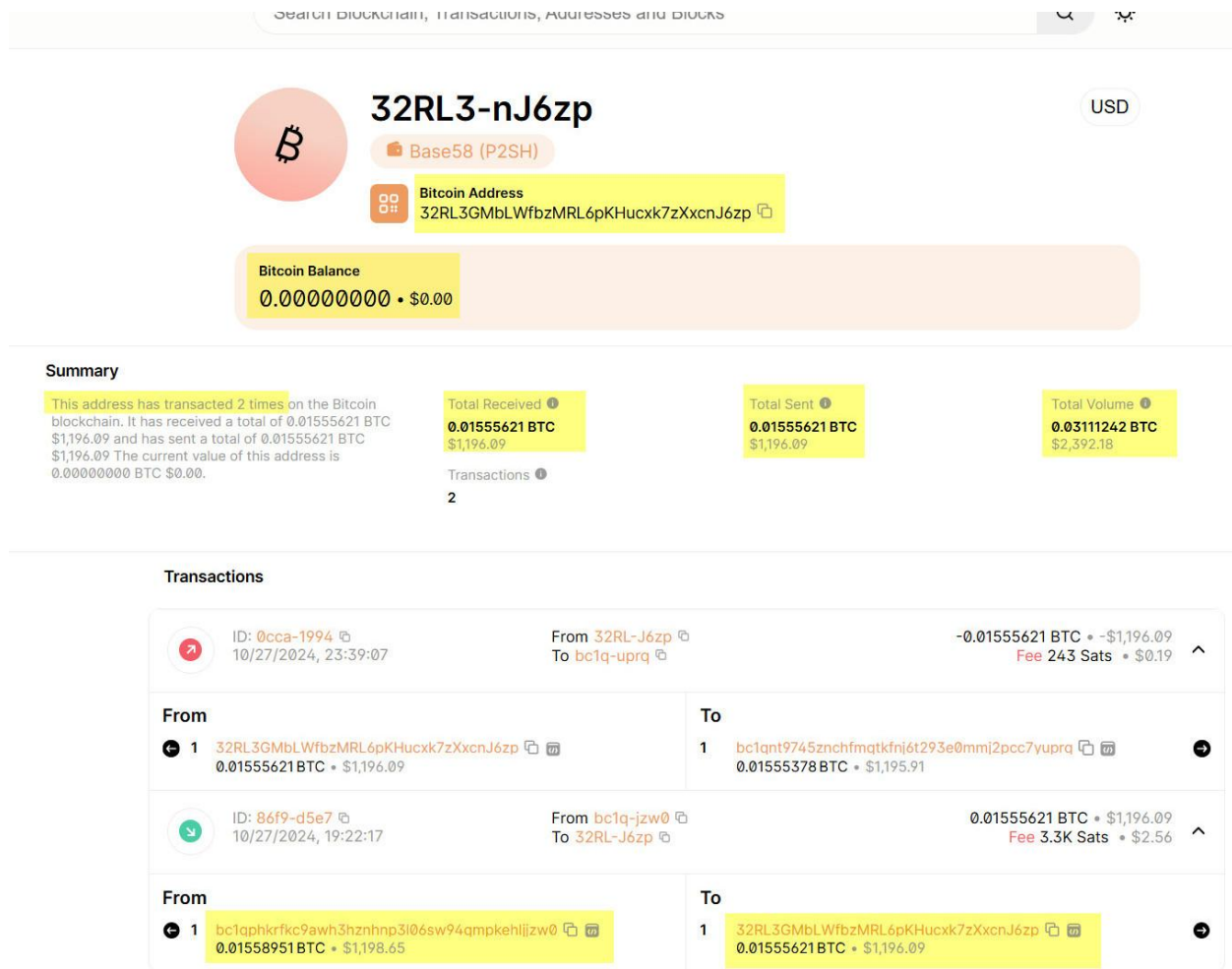


Figure 21: Reuse of downstream relay wallet
32RL3GMbLWfbzMRL6pKHucxk7zXxcnJ6zp confirming shared transaction pathways

5.4 Cross-Email Correlation Findings

The analysis identifies clear overlaps between wallet chains across multiple emails. The wallet **bc1qphkrfkc9awh3hznnp3i06sw94qmpkehljzw0** is reused in both Email 3 and Email 5 as a central aggregation node, while the wallet **32RL3GMbLWfbzMRL6pKHucxk7zXxcnJ6zp** appears as a downstream relay in the same two samples.

Additionally, the wallet **1EC7KpAFfCA9wbhq6MrpvHy557Yvcz3zqH** appears as an input contributor in a later aggregation transaction associated with email 4.

These overlaps demonstrate that the wallets are not isolated instances but components of a **shared and interconnected financial infrastructure**.

5.5 Behavioral Patterns Identified

The transaction analysis reveals several consistent behavioral patterns.

Funds are transferred rapidly across wallets, often within minutes to hours, with minimal delay between aggregation and forwarding. This strongly suggests automated or scripted transaction handling.

Transactions consistently exhibit fan-in aggregation, with 7 to 11 inputs merged into single outputs. This reflects consolidation of funds from multiple sources, corresponding to multiple victims.

In some cases, fan-out redistribution is observed, where funds are split across multiple outputs, including smaller-value transfers. This behavior suggests controlled redistribution and potential obfuscation.

All analyzed wallets maintain zero balance after transactions, indicating that they function exclusively as transit nodes rather than storage points.

Transaction amounts also show consistent ranges (approximately 0.008 to 0.02 BTC), suggesting standardized payment expectations rather than random transfers.

Finally, infrastructure reuse is clearly observed, with identical wallets and transaction patterns appearing across multiple emails, indicating a persistent and reusable laundering pipeline.

5.6 Assessment

The blockchain analysis demonstrates a structured and repeatable transaction workflow characterized by multi-input aggregation, rapid forwarding across chained wallets, reuse of intermediary nodes, and consistent transaction sizing.

These characteristics are consistent with a coordinated cryptocurrency handling infrastructure, rather than independent or opportunistic wallet usage.

5.7 Conclusion

The analyzed wallets form part of a linked transaction network in which funds are collected from multiple sources, aggregated into intermediary wallets, and rapidly forwarded through successive addresses before redistribution.

The presence of shared wallets across multiple email samples provides strong evidence of operational linkage between the investigated messages, confirming that they originate from the same underlying financial infrastructure.

6. Key Findings and Evidence Summary

6.1 Overview

The analysis of the collected email samples and associated infrastructure reveals a coordinated malicious email campaign consistent with phishing and sextortion tactics. The investigation identified a combination of technical indicators and behavioral patterns which, when analyzed collectively, provide strong evidence of organized and intentional malicious activity rather than isolated or opportunistic events.

6.2 Confirmed Technical Indicators

The investigation identified multiple technical indicators spanning network infrastructure, domain usage, and email configuration. The observed IP addresses are distributed across geographically diverse regions, including the United States, Singapore, and Malaysia, and are associated with cloud hosting providers such as DigitalOcean and Linode.

Domain analysis reveals a range of usage patterns, including privacy-protected domains with limited attribution, older domains potentially subject to subdomain abuse, spoofed legitimate domains, and, in some cases, non-existent or invalid domains used within return-path configurations. Additionally, several emails exhibit default or non-standard SMTP identifiers, such as “localhost” or generic hosting provider hostnames, indicating the use of minimally configured or deliberately manipulated mail systems.

Taken together, these indicators demonstrate consistent reliance on infrastructure that is non-attributable, easily replaceable, and operationally flexible.

6.3 Observed Attack Techniques

The campaign employs a combination of well-established techniques aligned with common phishing and extortion methodologies. Email spoofing is evident through the mismatch between displayed sender identities and the actual sending infrastructure, confirming the absence of legitimate domain control.

The use of VPS-based infrastructure enables attackers to deploy cloud-hosted servers for email delivery while minimizing attribution risks. Domain manipulation techniques are also prominent, including spoofing, subdomain abuse, and the use of unregistered or invalid domains to bypass traditional validation mechanisms.

Furthermore, the presence of direct SMTP abuse, particularly through the use of “localhost” HELO identifiers and non-standard configurations, indicates low-level email delivery practices designed to circumvent standard detection and traceability controls.

The variation of IP addresses, providers, and configurations across samples reflects an intentional strategy of infrastructure rotation.

6.4 Infrastructure Behavior Assessment

The infrastructure supporting the campaign exhibits distributed and heterogeneous architecture, characterized by inconsistent domain usage, variable identity configurations, and mixed reputation profiles across IP addresses. Both previously reported and newly deployed infrastructure components are observed, indicating a hybrid approach that balances reuse with rotation.

This operational model reflects a deliberate attempt to evade detection mechanisms, reduce correlation between campaign components, and maintain continuity even when individual elements are identified and blocked. The observed variability should therefore be interpreted as an intentional design feature rather than inconsistency.

The observed tactics and operational patterns are consistent with widely documented sextortion campaigns characterized by mass email distribution, standardied coercion templates, and cryptocurrency-based payment demands.

Such campaigns are typically conducted by financially motivated threat actors operating on a scale, often leveraging automation and rented infrastructure rather than advanced intrusion capabilities.

While direct attribution is not possible based on available evidence, the consistency of behavioral patterns, infrastructure usage, and financial workflows strongly supports the assessment of an organized and repeatable operational model aligned with known large-scale extortion activity.

6.5 Threat Classification

Based on the collected evidence, the activity can be classified as a phishing and sextortion email campaign leveraging rotating cloud-based infrastructure and multiple evasion techniques.

The combination of spoofed identities, infrastructure diversity, behavioral consistency, and coordinated financial activity aligns with known tactics used in financially motivated extortion schemes targeting individuals through large-scale email distribution.

6.6 Conclusion

The investigation confirms that the analyzed emails are part of a coordinated and structured malicious campaign. The convergence of technical indicators, infrastructure

patterns, delivery techniques, and financial behavior provides strong evidence of intentional abuse of cloud services to conduct scalable and evasive email-based attacks.

Rather than representing isolated incidents, the analyzed samples collectively demonstrate a unified operational model designed for persistence, adaptability, and mass targeting.

7. MITRE ATT&CK Mapping

7.1 Overview

The tactics, techniques, and procedures (TTPs) identified during this investigation can be systematically mapped to the MITRE ATT&CK framework. This mapping standardizes the observed adversarial behavior and aligns the findings with widely recognized threat intelligence models.

The analyzed campaign primarily operates across three core tactical domains: **Initial Access**, **Defense Evasion**, and **Impact**. Unlike traditional intrusion-based attacks, this activity does not rely on malware execution or system exploitation, but instead leverages social engineering combined with distributed and modular infrastructure to achieve its objectives.

7.2 Technique Mapping

1- Initial Access

- **Phishing –T1566**

The campaign relies entirely on large-scale email distribution as its primary entry vector. The messages are designed to simulate legitimacy and create urgency, serving as the mechanism through which victims are reached and manipulated. No alternative access vectors are observed.

2- Defense Evasion

- **Acquire Infrastructure – T1583**

The campaign demonstrates the use of externally acquired infrastructure, including cloud-based Virtual Private Servers (VPS), domains, and proxy services. These resources are low-cost, easily replaceable, and enable rapid scaling of operations while minimizing attribution.

- **Proxy – T1090**

The use of intermediary services such as Cloudflare introduces an abstraction layer between the attacker and the underlying infrastructure. This reduces visibility into the origin systems and complicates attribution efforts.

- **Infrastructure Rotation and Indicator Variability (Behavioral Mapping)**

While not mapped to a single explicit technique, the observed rotation of IP addresses, domains, and configurations reflects a deliberate evasion strategy. This

behavior reduces the effectiveness of static detection mechanisms and limits long-term tracking of campaign components.

- **Email Identity Spoofing (Behavioral Mapping within Phishing)**

The mismatch between displayed sender identities and actual sending infrastructure demonstrates deliberate identity manipulation. This technique enhances credibility and increases the likelihood of successful victim engagement.

3- Impact

- **Impact (Custom Mapping: Cryptocurrency-Based Extortion)**

The primary objective of the campaign is to coerce victims into transferring cryptocurrency under the threat of reputational harm. The observed Bitcoin wallet chains and transaction patterns indicate a structured financial extraction process, with funds rapidly aggregated and forwarded through intermediary wallets.

7.3 Analytical Insight

The MITRE ATT&CK mapping highlights that the campaign is fundamentally **behavior-driven rather than exploit-driven**. The absence of malware, persistence mechanisms, or command-and-control infrastructure distinguishes this activity from traditional intrusion campaigns.

Instead, the attacker achieves effectiveness through:

- Scalable phishing distribution
- Identity spoofing and infrastructure obfuscation
- Structured and reusable financial transaction workflows

This classification underscores the importance of shifting detection strategies from signature-based approaches toward **behavioral and pattern-based analysis**, particularly for threats that rely on psychological manipulation rather than technical compromise.

Furthermore, the combination of infrastructure acquisition, rapid rotation, and financial pipeline reuse reflects a coordinated operational model aligned with known large-scale extortion campaigns.

8. Cyber Kill Chain Analysis

8.1 Overview

The investigated campaign can be effectively analyzed using the Cyber Kill Chain framework to illustrate the sequential stages of the attack lifecycle. This model provides a structured perspective on how the threat actor progresses from initial targeting to the achievement of operational objectives.

Unlike traditional cyber intrusions, this campaign does not involve malware deployment or system compromise. Instead, it adapts the kill chain by replacing technical exploitation stages with psychological manipulation, while still maintaining a structured and repeatable attack flow.

8.2 Attack Lifecycle Analysis

Reconnaissance

The attacker targets a broad and non-specific set of email addresses, likely obtained through publicly available data, prior breaches, or automated harvesting techniques. No evidence of targeted reconnaissance or victim profiling is observed, indicating a mass-distribution approach.

Weaponization

A standardized sextortion email template is prepared, incorporating psychological triggers such as fear, urgency, and reputational threat. The message structure is consistent across all samples, suggesting automation and pre-defined content generation rather than manual crafting.

Delivery

The malicious emails are distributed through cloud-based SMTP infrastructure leveraging Virtual Private Servers (VPS), spoofed sender identities, and diverse domain configurations. The use of distributed infrastructure enhances delivery success while reducing the likelihood of detection and blocking.

Exploitation

No software vulnerability is exploited. Instead, the campaign relies entirely on psychological coercion to manipulate victims. The attacker attempts to induce panic and urgency, compelling recipients to act without verification or critical assessment.

Installation

No malware installation or persistence mechanisms were observed. The campaign does not introduce any executable payload or establish a foothold on the victim's system.

Command and Control (C2)

No persistent command-and-control channels were identified. Interaction is limited to one-way email communication, with no evidence of remote access or ongoing communication beyond the initial message.

Actions on Objectives

The attacker achieves the intended objective when victims transfer Bitcoin to the provided wallets. This initiates the structured transaction workflow identified in the blockchain analysis, enabling fund aggregation and redistribution through intermediary wallets.

8.3 Analytical Insight

The application of the Cyber Kill Chain to this campaign demonstrates a **non-traditional attack model**, where technical intrusion phases are replaced by human-centric exploitation. The absence of malware, persistence, and command-and-control infrastructure highlights a shift toward low-complexity, high-scalability threat operations.

This model reinforces that:

- The **primary attack surface is the human factor**, not the technical environment
- The **exploitation phase is psychological rather than technical**
- The **success of the attack depends on behavioral response rather than system vulnerability**

Consequently, effective detection and mitigation require a shift from conventional intrusion-focused defenses toward **user awareness, behavioral analysis, and email-based threat detection strategies**.

Furthermore, the clear alignment between delivery mechanisms and financial outcomes demonstrates a streamlined operational workflow, where each stage directly contributes to rapid and scalable monetization.

9. Diamond Model Analysis

9.1 Overview

The investigated campaign can be analyzed using the Diamond Model of Intrusion Analysis, which provides a structured framework for correlating the four core elements of an intrusion: **Adversary, Infrastructure, Capability, and Victim**.

This model is particularly suitable for the current investigation, as it enables the integration of email forensics, infrastructure analysis, and blockchain transaction tracing into a unified analytical structure. By examining the relationships between these components, it becomes possible to assess operational consistency and infer potential linkage across multiple observed samples.

9.2 Diamond Components

Adversary

The threat actor remains unidentified; however, the observed behavior indicates a moderately organized and operationally consistent actor. The campaign demonstrates the ability to conduct large-scale email distribution, operate distributed infrastructure, and manage a structured cryptocurrency transaction workflow. The consistency of messaging structure, infrastructure usage patterns, and financial behavior across all samples suggests a single coordinated actor or a tightly linked operational group, rather than independent or opportunistic activity.

Infrastructure

The campaign relies on a distributed and modular infrastructure composed of:

- Cloud-based Virtual Private Servers (VPS) hosted across multiple geographic regions
- Diverse domain configurations, including disposable domains, spoofed identities, and potentially abused subdomains
- Proxy and intermediary services (e.g., Cloudflare) used to obscure origin infrastructure
- A structured network of Bitcoin wallets forming a multi-stage transaction pipeline

The infrastructure is characterized by high replaceability, geographic dispersion, and abstraction layers, enabling operational resilience and reducing attribution visibility.

Capability

The actor demonstrates capabilities centered on social engineering and operational scalability, including:

- Mass email distribution using SMTP-based delivery mechanisms
- Email identity spoofing and domain manipulation
- Use of minimally configured or misconfigured mail infrastructure
- Psychological coercion techniques leveraging fear, urgency, and reputational threat
- Coordinated cryptocurrency handling through aggregation and relay wallet structures

No evidence of malware deployment, exploitation of software vulnerabilities, or persistence mechanisms was observed, reinforcing the non-intrusive nature of the campaign.

Victim

The targets consist of general email users with no indication of specific profiling or targeted selection. Victims are selected opportunistically through mass distribution, with minimal personalization limited to the recipient's email address. The effectiveness of the campaign depends on psychological pressure and behavioral response rather than technical compromise, making any email recipient a potential target.

9.3 Relationship Analysis

The Diamond Model highlights consistent and reinforcing relationships between all components:

- The **Adversary** leverages scalable **Capabilities** (phishing and coercion)
- These capabilities are delivered through adaptable **Infrastructure** (VPS, domains, SMTP services)
- The **Victim** is engaged through psychological manipulation rather than technical exploitation
- Financial interaction is facilitated through cryptocurrency wallets, linking victim response to adversary-controlled infrastructure

A key observation is the linkage between infrastructure and financial components. The reuse of intermediary Bitcoin wallets across multiple email samples provides strong evidence of operational continuity and shared transactional infrastructure rather than isolated financial activity.

9.4 Analytical Insight

The application of the Diamond Model indicates that the campaign operates as a cohesive and coordinated system in which each component reinforces the others to achieve scalability and operational persistence.

Key insights include:

- The infrastructure is modular and replaceable, supporting continuous operation despite defensive disruption
- The capability is primarily behavioral, relying on psychological manipulation rather than technical exploitation
- The financial workflow serves as a key correlation layer, linking multiple campaign instances through shared transaction paths
- The lack of victim targeting specificity supports a high-volume, opportunistic distribution strategy

Overall, the model demonstrates that the campaign's effectiveness stems from the integration of social engineering, infrastructure abstraction, and coordinated financial processing rather than traditional intrusion-based techniques.

10. Indicators of Compromise (IOC) – SOC-Ready Table

10.1 Overview

This section consolidates all identified Indicators of Compromise (IOCs) observed during the investigation. The indicators cover network infrastructure, email artifacts, domain usage patterns, and cryptocurrency-related entities associated with the campaign.

These artifacts are intended to support SOC operations, including detection engineering, threat hunting, correlation analysis, and incident response enrichment. It is important to note that several indicators are volatile and may change rapidly due to the attacker's infrastructure rotation strategy.

10.2 IOC Table

Network / Infrastructure Indicators

Type	Indicator	Description
IP Infrastructure	VPS-hosted infrastructure (DigitalOcean, Linode, Malaysia-based providers)	Cloud-based SMTP delivery infrastructure used for campaign distribution across multiple regions
Hosting Environment	Cloud-based Virtual Private Server (VPS) environments	Disposable and scalable infrastructure enabling rapid rotation and redeployment

Domain Indicators

Type	Indicator	Description
Domain Type	Disposable domains	Temporarily registered domains used for short-lived malicious activity
Domain Type	Spoofed legitimate domains	Domains impersonating trusted entities to increase credibility
Domain Type	Abused subdomains	Potentially compromised or misused subdomain configurations
Domain Status	Invalid or unregistered return-path domains	Indicates malformed or non-standard email authentication setup
Infrastructure Obfuscation	Reverse proxy services (e.g., Cloudflare)	Used to conceal origin servers and complicate attribution analysis

Email Indicators

Type	Indicator	Description
SMTP HELO Identifier	"localhost"	Indicative of misconfigured or directly abused SMTP services
SMTP Hostname	Generic hosting provider identifiers	Suggests default or minimally configured mail infrastructure
Sender Identity	Mismatch between displayed and actual sending infrastructure	Clear evidence of email spoofing and identity manipulation
Message Template	Standardized sextortion email structure	Repetitive psychological coercion-based messaging pattern used across samples

Cryptocurrency Indicators

Type	Indicator	Description
Wallet Address	bc1qphkrfk9awh3hznnp3l06sw94qmpkehljjzw0	Reused aggregation wallet observed across multiple email campaigns
Wallet Address	32RL3GMbLWfbzMRL6pKHucxk7zXxcnJ6zp	Shared relay wallet used across multiple transaction chains
Transaction Behavior	Multi-input aggregation	Consolidation of funds from multiple independent sources into a single transaction
Transaction Behavior	Immediate fund forwarding	Rapid transfer of received funds with no storage or retention
Transaction Pattern	Fan-in / Fan-out structure	Structured redistribution pattern consistent with laundering-style workflows

10.3 Analytical Note

While the identified indicators provide value for detection and correlation, many are inherently volatile (e.g., IP addresses, domains, and hosting infrastructure) and can be rapidly replaced by the adversary as part of an infrastructure rotation strategy.

Accordingly, effective SOC detection and threat hunting should prioritize:

- Behavioral patterns in email structure and coercion language
- Infrastructure relationship mapping and reuse patterns
- Cryptocurrency transaction flows and wallet interconnections

rather than relying solely on static indicators, which offer limited long-term detection value.

These indicators should be correlated with email authentication results (SPF, DKIM, DMARC failures), SMTP anomalies, and SIEM behavioral alerts to enable effective detection, enrichment, and investigation workflows.

11. Pyramid of Pain Analysis

11.1 Overview

The Pyramid of Pain model is used to evaluate the defensive value of different categories of indicators based on the level of effort required for an adversary to modify or replace them. This framework enables prioritization of detection strategies that maximize long-term disruption of adversary operations by focusing on indicators that impose higher operational cost on the threat actor.

In the context of this investigation, the model is applied to assess which observed indicators can be reliably blocked and which should be prioritized for behavioral and intelligence-driven detection.

11.2 Indicator Classification

Low Pain (Easy to Change)

- IP addresses (rotated through VPS infrastructure across cloud providers)
- Domain names (disposable, newly registered, or rapidly abandoned domains)
- SMTP server identifiers and mail server hostnames

These indicators are highly volatile and can be replaced with minimal operational effort and cost. They provide limited long-term detection value and are primarily useful for immediate blocking, enrichment, and short-term correlation.

Medium Pain

- Hosting providers and infrastructure vendors (e.g., DigitalOcean, Linode)
- Proxy and obfuscation services (e.g., Cloudflare usage patterns)
- Email infrastructure configuration anomalies (e.g., inconsistent, or non-standard SMTP setups)

These indicators require moderate effort to modify due to reliance on third-party services and infrastructure dependencies. However, they remain relatively easy for adversaries to replicate by switching providers or adjusting configurations.

High Pain

- Email content structure consistency (standardized sextortion templates)

- Psychological coercion patterns (fear induction, urgency, reputational threats)
- Cryptocurrency transaction behavior patterns (fan-in / fan-out fund flows)
- Reuse of intermediary cryptocurrency wallets across multiple campaigns

These indicators reflect adversary operational behavior rather than static infrastructure. Disruption at this level imposes significant operational cost by requiring changes to campaign design, messaging strategy, and financial handling processes.

TTP-Level Indicators (Highest Value Layer)

- Structured cryptocurrency laundering workflows (multi-stage wallet chaining architecture)
- Reused financial infrastructure nodes across multiple campaigns
- End-to-end attack workflow (phishing delivery → psychological coercion → payment extraction → laundering pipeline)

These indicators represent core adversary tactics, techniques, and procedures (TTPs). Detection and disruption at this level have the greatest defensive impact, directly affecting campaign scalability, financial efficiency, and operational continuity.

11.3 Analytical Insight

The analysis demonstrates that reliance on low-level indicators such as IP addresses and domain names provides limited long-term defensive value due to their high volatility and ease of replacement by the adversary.

Greater defensive effectiveness is achieved by shifting focus toward:

- Behavioral patterns in phishing and coercion messaging
- Structured cryptocurrency transaction flows and wallet reuse relationships
- Reusable operational and infrastructure interaction patterns across campaigns

This approach transitions detection strategy from reactive indicator-based blocking to proactive, intelligence-driven disruption of adversary workflows, financial pipelines, and operational procedures.

12. Recommendations & Mitigation Strategies

12.1 Overview

Based on the findings of this investigation, the analyzed campaign relies on large-scale email distribution, identity spoofing, psychological coercion, and structured cryptocurrency workflows rather than technical exploitation. Accordingly, effective mitigation requires a combination of **email security controls, user awareness, behavioral detection, and threat intelligence integration**.

The following recommendations are designed to support **Security Operations Center (SOC) teams, organizations, and individual users** in preventing, detecting, and responding to similar campaigns.

12.2 Email Security Controls

Organizations should strengthen email authentication and filtering mechanisms to reduce exposure to spoofed and malicious messages.

- Enforce strict implementation of **SPF, DKIM, and DMARC policies** with a “reject” enforcement level to prevent domain spoofing
- Deploy advanced email security gateways capable of detecting phishing and social engineering patterns
- Implement anomaly-based filtering to identify unusual sender behavior, mismatched headers, and non-standard SMTP configurations
- Block or flag emails originating from known VPS hosting ranges or low-reputation infrastructure when appropriate
- Monitor for malformed or invalid return-path domains and suspicious HELO identifiers (e.g., “localhost”)

These controls reduce the likelihood of successful delivery and increase early detection of malicious emails.

12.3 Security Awareness and User Training

As the campaign relies heavily on psychological manipulation, user awareness is a critical defensive layer.

- Conduct regular security awareness training focused on phishing and sextortion scenarios

- Educate users to recognize common coercion techniques, including urgency, fear-based messaging, and threats of data exposure
- Reinforce the principle that unsolicited claims of device compromise should not be trusted without verification
- Encourage users to report suspicious emails to the security team rather than engaging with the sender
- Promote a culture of verification and critical assessment before taking any action involving financial transactions

Improving user awareness directly reduces the effectiveness of social engineering-based attacks.

12.4 Detection and Monitoring (SOC Focus)

SOC teams should prioritize behavioral detection and correlation over reliance on static indicators.

- Develop detection rules for:
 - Repetitive sextortion email templates and coercive language patterns
 - Mismatch between sender identity and originating infrastructure
 - SMTP anomalies, including generic HELO identifiers and misconfigured mail servers
- Correlate email events with:
 - Authentication failures (SPF, DKIM, DMARC)
 - Known phishing patterns and historical campaign data
- Implement SIEM use cases that identify high-volume email distribution from cloud-based infrastructure
- Monitor for recurring infrastructure patterns across multiple alerts, even when individual indicators differ

Behavioral detection enables identification of campaigns despite infrastructure rotation.

12.5 Threat Intelligence Integration

Integrating threat intelligence enhances both detection and investigation capabilities.

- Enrich email and network indicators using external intelligence sources such as AbuseIPDB

- Track infrastructure reuse patterns across campaigns, including IP ranges, hosting providers, and domain behaviors
- Monitor for recurring cryptocurrency wallets and transaction patterns associated with extortion activity
- Share relevant indicators and findings with internal teams and, where appropriate, external intelligence-sharing communities
- Map observed activity to frameworks such as MITRE ATT&CK to standardize detection and reporting

This approach improves contextual awareness and supports proactive threat hunting.

12.6 Cryptocurrency Monitoring and Response

Given the financial objective of the campaign, organizations should incorporate visibility into cryptocurrency-related activity.

- Monitor and flag known malicious or suspicious wallet addresses identified during investigations
- Analyze transaction patterns such as rapid fund forwarding, multi-input aggregation, and wallet reuse
- Where possible, collaborate with blockchain analysis platforms and law enforcement for tracking illicit financial flows
- Educate users that legitimate organizations do not request payments via cryptocurrency for security-related issues

Although prevention at this stage is limited, visibility into financial flows supports investigation and attribution efforts.

12.7 Incident Response Procedures

Organizations should define clear procedures for handling sextortion and phishing-related incidents.

- Establish a reporting mechanism for users to submit suspicious emails
- Isolate and analyze reported messages, including header and infrastructure examination
- Block identified indicators across email and network security controls
- Conduct internal awareness notifications if multiple users are targeted

- Document and retain evidence for potential legal or intelligence purposes

A structured response process ensures rapid containment and consistent handling of incidents.

12.8 Strategic Defensive Approach

The findings of this investigation highlight that traditional, indicator-based defenses are insufficient against highly adaptable campaigns.

Effective long-term mitigation should prioritize:

- Behavioral and pattern-based detection
- Intelligence-driven security operations
- Continuous monitoring of infrastructure and financial workflows
- Integration of human-focused defenses alongside technical controls

By shifting from reactive blocking to proactive, intelligence-led defense strategies, organizations can significantly reduce the impact and scalability of similar campaigns.

13. Final Conclusion

13.1 Summary of Findings

This investigation analyzed five distinct email samples associated with a sextortion campaign leveraging spoofed identities, distributed infrastructure, and cryptocurrency-based payment mechanisms. The combined analysis of email content, header artifacts, infrastructure components, and blockchain transactions confirms that the activity represents a coordinated, large-scale, and financially motivated operation.

Despite variation in technical indicators such as IP addresses, domains, and individual wallet identifiers, strong consistency is observed across:

- Email structure and coercion techniques
- Infrastructure deployment patterns
- Cryptocurrency transaction workflows

These correlations provide high confidence that the analyzed samples are part of a single operational campaign or closely related activity set, rather than independent incidents.

13.2 Nature of the Threat

The campaign is characterized by a non-intrusive, behavior-driven attack model. No evidence of malware deployment, system compromise, persistence mechanisms, or command-and-control infrastructure was identified throughout the investigation.

Instead, the attacker relies entirely on:

- Large-scale phishing distribution
- Identity spoofing and infrastructure abstraction
- Psychological coercion leveraging fear, urgency, and reputational threats

This confirms that the primary attack vector is human behavior rather than technical vulnerability, representing a shift toward low-complexity, high-scalability cyber-enabled fraud operations.

13.3 Operational Assessment

The infrastructure supporting the campaign demonstrates a modular, distributed, and highly adaptable design, including:

- Cloud-based Virtual Private Servers (VPS) across multiple geographic regions

- Disposable and spoofed domain configurations
- Proxy-based obfuscation layers
- Flexible SMTP delivery mechanisms

In parallel, blockchain analysis reveals a structured and repeatable financial workflow, characterized by:

- Multi-input aggregation of funds (fan-in)
- Rapid forwarding across chained wallets
- Reuse of intermediary wallets across multiple emails
- Absence of balance retention at all transaction stages

This combination of infrastructure flexibility and financial structuring indicates a moderately sophisticated operational model designed for scalability, resilience, and evasion.

13.4 Intelligence and Attribution Insight

While direct attribution to a specific individual or group is not possible based on available data, the consistency of tactics, infrastructure usage, and financial behavior strongly suggests:

- A single coordinated threat actor, or
- A closely aligned operational group utilizing shared resources and workflows

The reuse of cryptocurrency wallets across multiple email samples provides a key correlation point, reinforcing the existence of a persistent and reusable financial infrastructure underlying the campaign.

13.5 Defensive Implications

The findings of this investigation highlight several critical implications for defensive strategy:

- Traditional indicator-based detection is insufficient due to rapid infrastructure rotation
- Behavioral patterns and adversary workflows provide more reliable detection opportunities
- The human element represents the primary point of exploitation and must be addressed through awareness and training

- Financial transaction analysis can serve as a valuable correlation and intelligence layer

Effective defense against similar campaigns requires a transition toward intelligence-driven and behavior-focused security operations, rather than reliance on static indicators alone.

13.6 Final Assessment

The analyzed campaign represents a scalable, low-complexity, and high-impact cyber-enabled extortion operation that leverages social engineering, distributed infrastructure, and structured cryptocurrency handling to achieve its objectives.

The investigation demonstrates that even in the absence of technical compromise, a comprehensive approach combining:

- Email forensics
- Infrastructure analysis
- Blockchain tracing
- Threat intelligence modeling

enables the identification, correlation, and understanding of adversary operations at a high level of confidence.

This case underscores the evolving nature of modern cyber threats, where operational efficiency and psychological manipulation can replace technical sophistication while still achieving significant impact.

14. References

14.1 Investigative Data Sources (Per Email Analysis)

The following sources were used to validate infrastructure, domain registration data, IP reputation, and cryptocurrency transaction flows for each analyzed email sample.

Email 1 – Source References

- IP Intelligence: IPinfo – <https://ipinfo.io/67.207.81.68>
- IP Reputation: AbuseIPDB – <https://www.abuseipdb.com/check/67.207.81.68>
- Domain WHOIS (Return-Path): WHOIS – <https://www.whois.com/whois/cedarchic.com>
- Domain WHOIS (HELO): WHOIS – <https://www.whois.com/whois/sppcongresos.com>
- Blockchain Analysis: Blockchain.com
 - <https://www.blockchain.com/explorer/addresses/btc/1EC7KpAFfCA9wbhq6MrpvHy557Yvcz3zqH>
 - <https://www.blockchain.com/explorer/addresses/BTC/bc1q4g37wkvgppcg5qru8gudnqf3qjqcej4a966ct8>
 - <https://www.blockchain.com/explorer/addresses/BTC/bc1qcpwhs4nj30zvnx7x8cyt9n3996easp58shuuc2>

Email 2 – Source References

- IP Intelligence: IPinfo – <https://ipinfo.io/198.211.105.148>
- IP Reputation: AbuseIPDB – <https://www.abuseipdb.com/check/198.211.105.148>
- Domain WHOIS: WHOIS – <https://www.whois.com/whois/enrgio.com>
- Blockchain Analysis: Blockchain.com
 - <https://www.blockchain.com/explorer/addresses/btc/1CWTFeMfPCG1Q6uVLSpHUmQ1J1i6hxj1LK>
 - <https://www.blockchain.com/explorer/addresses/BTC/bc1q68jcwyclqtmv evguse9h8z30lu23fy9dplkd8>
 - <https://www.blockchain.com/explorer/addresses/BTC/bc1qd4lssfg5dl5sfc m4qa2q5trvflksyav8jqpeaq>

Email 3 – Source References

- IP Intelligence: IPinfo – <https://ipinfo.io/202.59.9.32>
- IP Reputation: AbuseIPDB – <https://www.abuseipdb.com/check/202.59.9.32>
- Domain WHOIS: WHOIS – <https://www.whois.com/whois/eziakaun.com>
- Blockchain Analysis: Blockchain.com
 - <https://www.blockchain.com/explorer/addresses/btc/19tWNhBbjDHBHk9FqQzhU5k7w2E9pRQrrU>
 - <https://www.blockchain.com/explorer/addresses/BTC/bc1qphkrkfc9awh3hznhrp3l06sw94qmpkehljzw0>
 - <https://www.blockchain.com/explorer/addresses/BTC/32RL3GMbLWfbzMRL6pKHucxk7zXxcnJ6zp>

Email 4 – Source References

- IP Intelligence: IPinfo – <https://ipinfo.io/152.42.239.74>
- IP Reputation: AbuseIPDB – <https://www.abuseipdb.com/check/152.42.239.74>
- Domain WHOIS: WHOIS – <https://www.whois.com/whois/cggl.com>
- Blockchain Analysis: Blockchain.com
 - <https://www.blockchain.com/explorer/addresses/btc/1Hf9QGAtsXHe89fLZji99c6iDrnqHkcyGq>
 - <https://www.blockchain.com/explorer/addresses/BTC/bc1qcvw86544a6n9liy4gjykh8dud3nk4udl8f30vz>
 - <https://www.blockchain.com/explorer/addresses/BTC/bc1qyv542p6tu3gh7cfkl8mw7c3ft5penppjyysh6>

Email 5 – Source References

- IP Intelligence: IPinfo – <https://ipinfo.io/45.79.252.40>
- IP Reputation: AbuseIPDB – <https://www.abuseipdb.com/check/45.79.252.40>
- Domain WHOIS (Return-Path): WHOIS – <https://www.whois.com/whois/casainterativa.com>

- Domain Reference (HELO/Registration): WHOIS – <https://www.whois.com/whois/linodeusercontent.com>
- Blockchain Analysis: Blockchain.com
 - <https://www.blockchain.com/explorer/addresses/btc/1L3Nd5uG7ZVxXkd3SvivjZk4crVPAKNqQA>
 - <https://www.blockchain.com/explorer/addresses/BTC/bc1qphkrfkc9awh3hznhnp3l06sw94qmpkehljjzw0>
 - <https://www.blockchain.com/explorer/addresses/BTC/32RL3GMbLWfbzMRL6pKHucxk7zXxcnJ6zp>

14.2 Analytical Frameworks and Methodologies

The investigation established cybersecurity and threat intelligence frameworks to structure analysis and ensure methodological rigor:

- MITRE ATT&CK – <https://attack.mitre.org/>
- Lockheed Martin Cyber Kill Chain – <https://www.lockheedmartin.com/en-us/capabilities/cyber/cyber-kill-chain.html>
- Diamond Model of Intrusion Analysis – <https://www.activeresponse.org/wp-content/uploads/2013/07/diamond.pdf>
- SANS Institute – Pyramid of Pain model

14.3 Supporting Intelligence and Analysis Platforms

The following platforms supported correlation, validation, and enrichment throughout the investigation:

- IPinfo – IP geolocation and ASN analysis
- AbuseIPDB – Reputation and abuse reporting
- WHOIS – Domain ownership and registration data
- Blockchain.com – Transaction tracing and wallet analysis

14.4 Analytical Note

All references included in this section correspond to:

- Direct investigative sources used for evidence collection and validation

- External platforms used for enrichment and correlation
- Standardized frameworks applied for threat modeling and analysis

These references ensure traceability, reproducibility, and analytical transparency across all findings presented in this report.

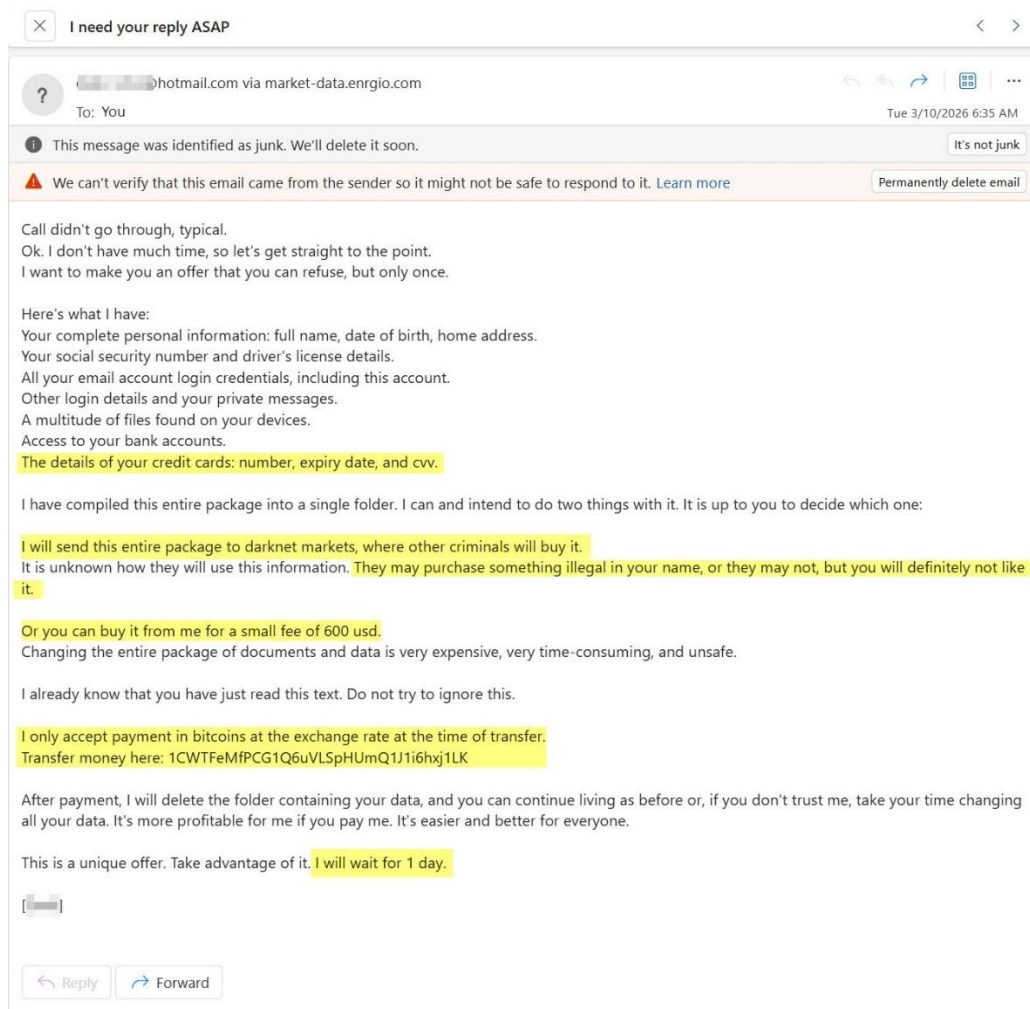
Appendix A: Phishing Email Content (Emails 2–5)

This appendix contains:

- Screenshots of email bodies
- Highlighted extortion and threat language
- Payment instructions and wallet references

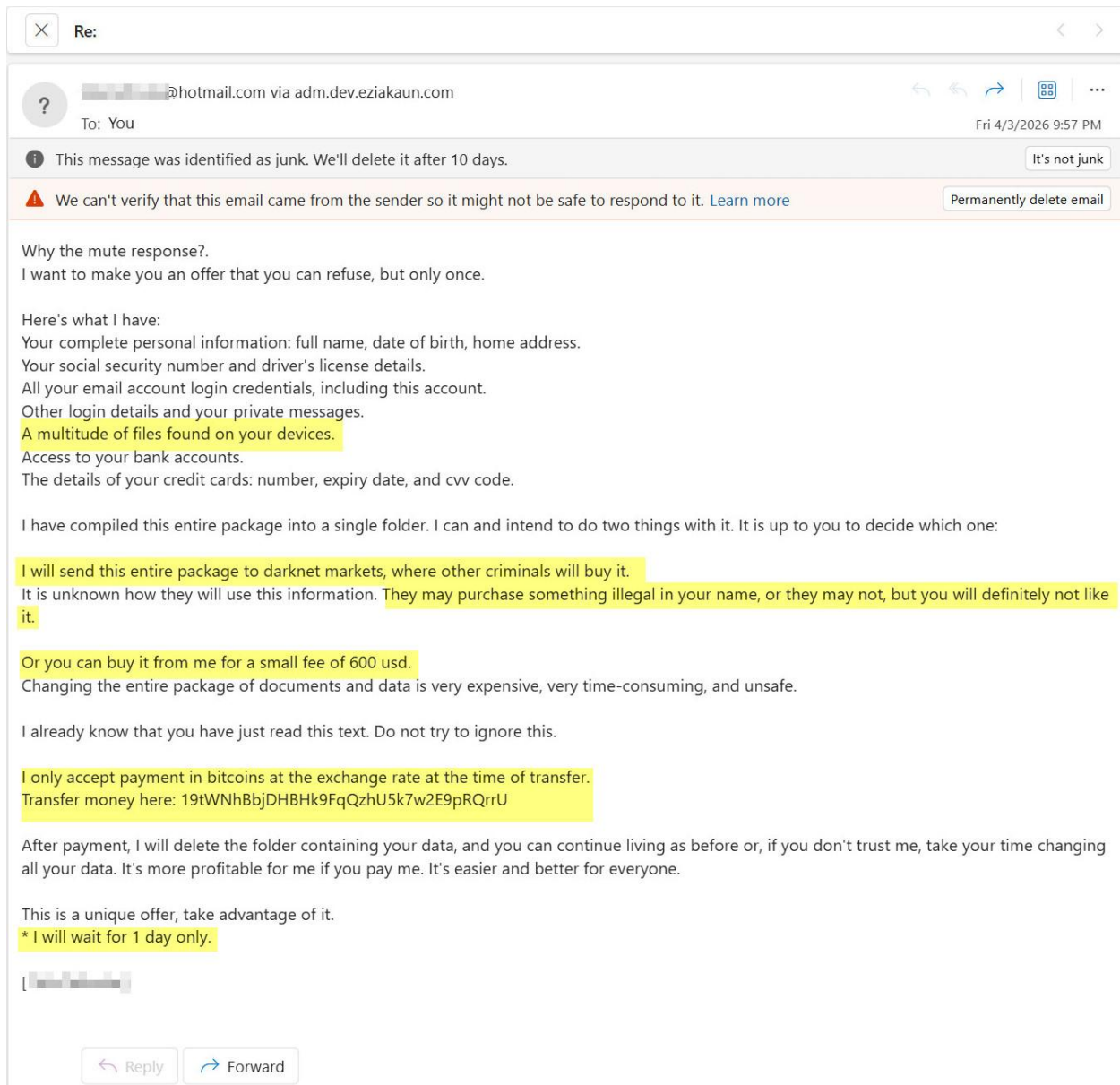
Linked to section: 3. Email Content Analysis

Figure A1: Email 2 content



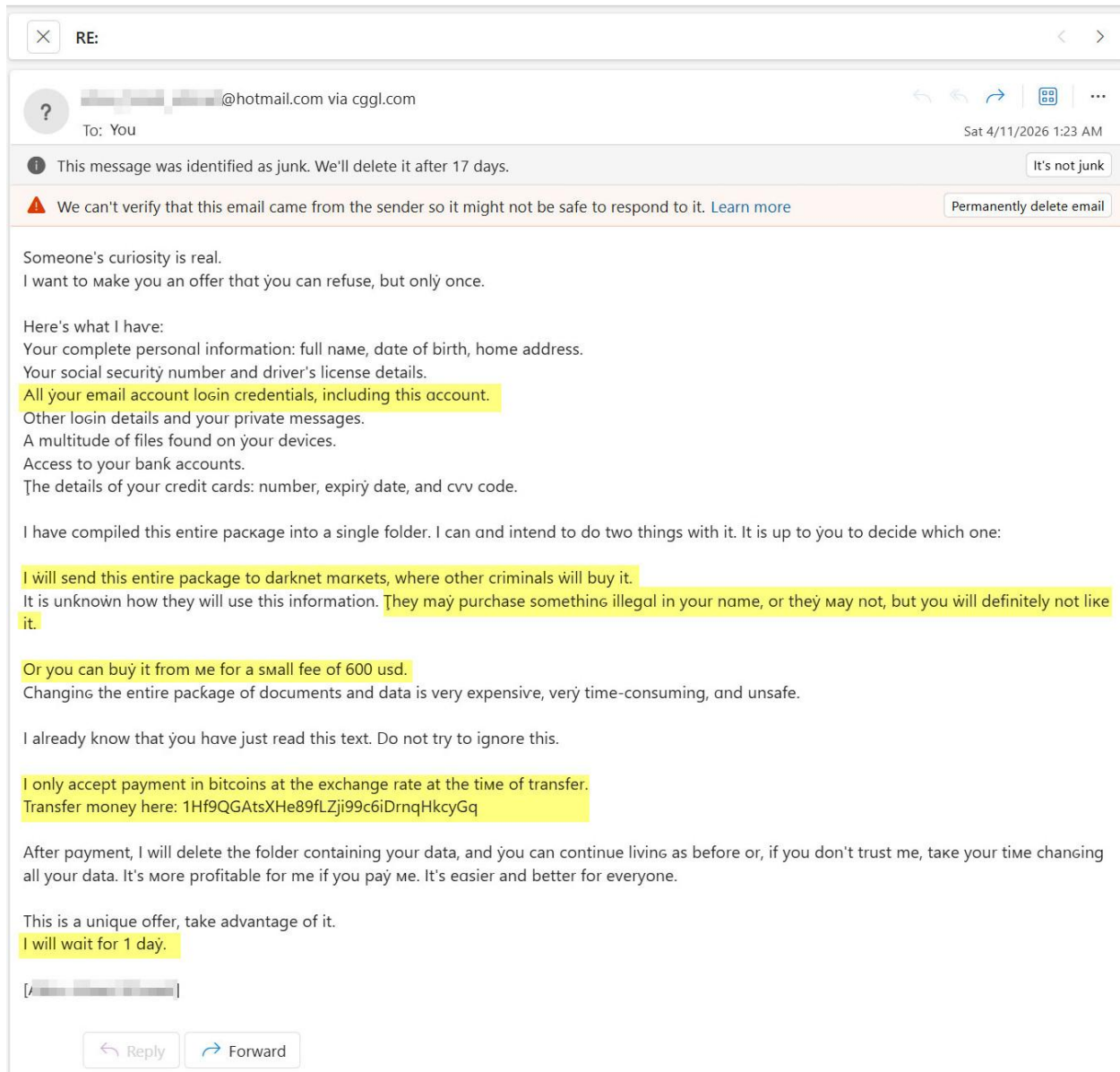
Sextortion email excerpt highlighting psychological coercion, financial demand, and Bitcoin-based payment instructions.

Figure A2: Email 3 content



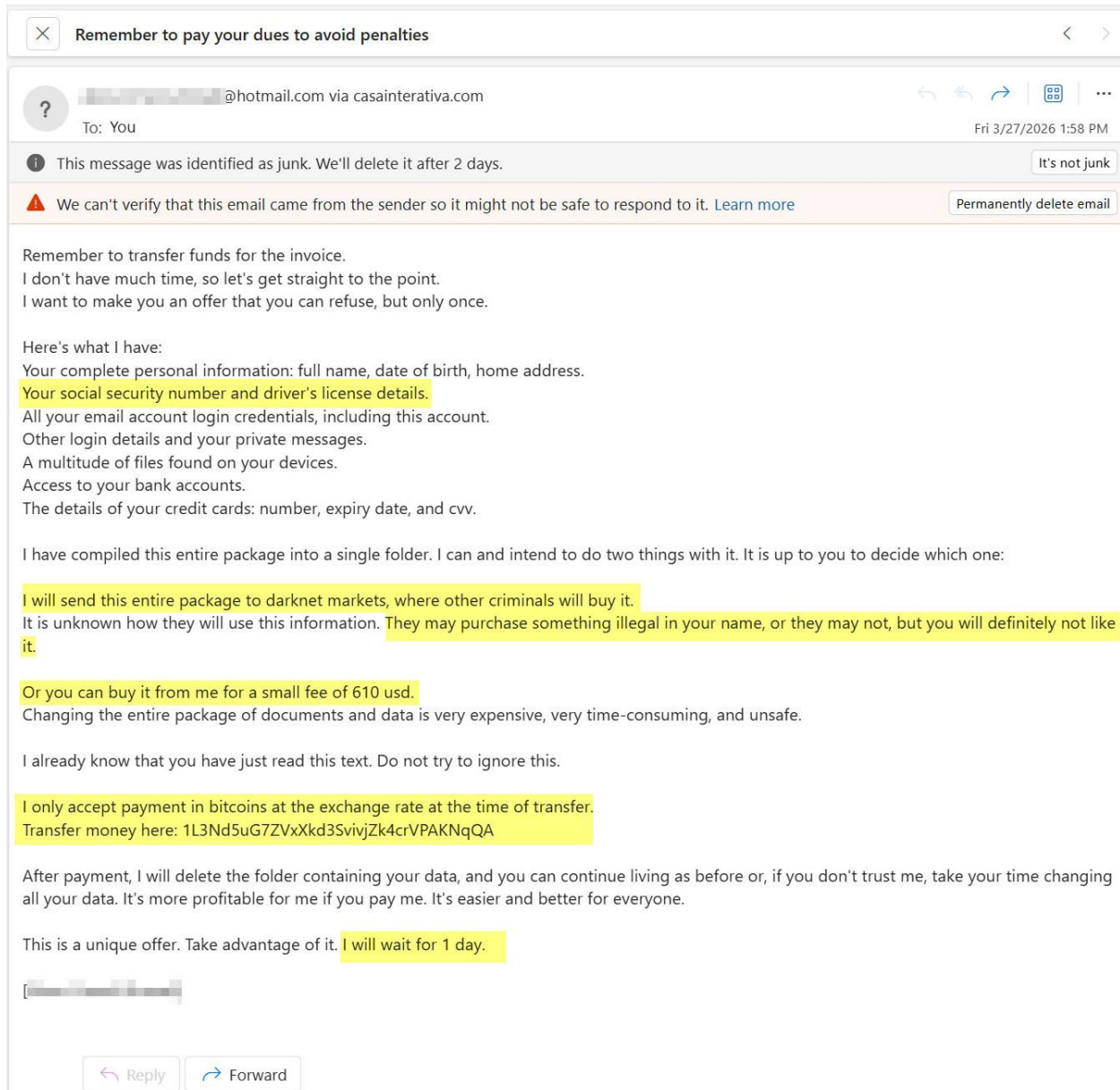
Sextortion email excerpt highlighting psychological coercion, financial demand, and Bitcoin-based payment instructions.

Figure A3: Email 4 content



Sextortion email excerpt highlighting psychological coercion, financial demand, and Bitcoin-based payment instructions.

Figure A4: Email 5 content



Sextortion email excerpt highlighting psychological coercion, financial demand, and Bitcoin-based payment instructions.

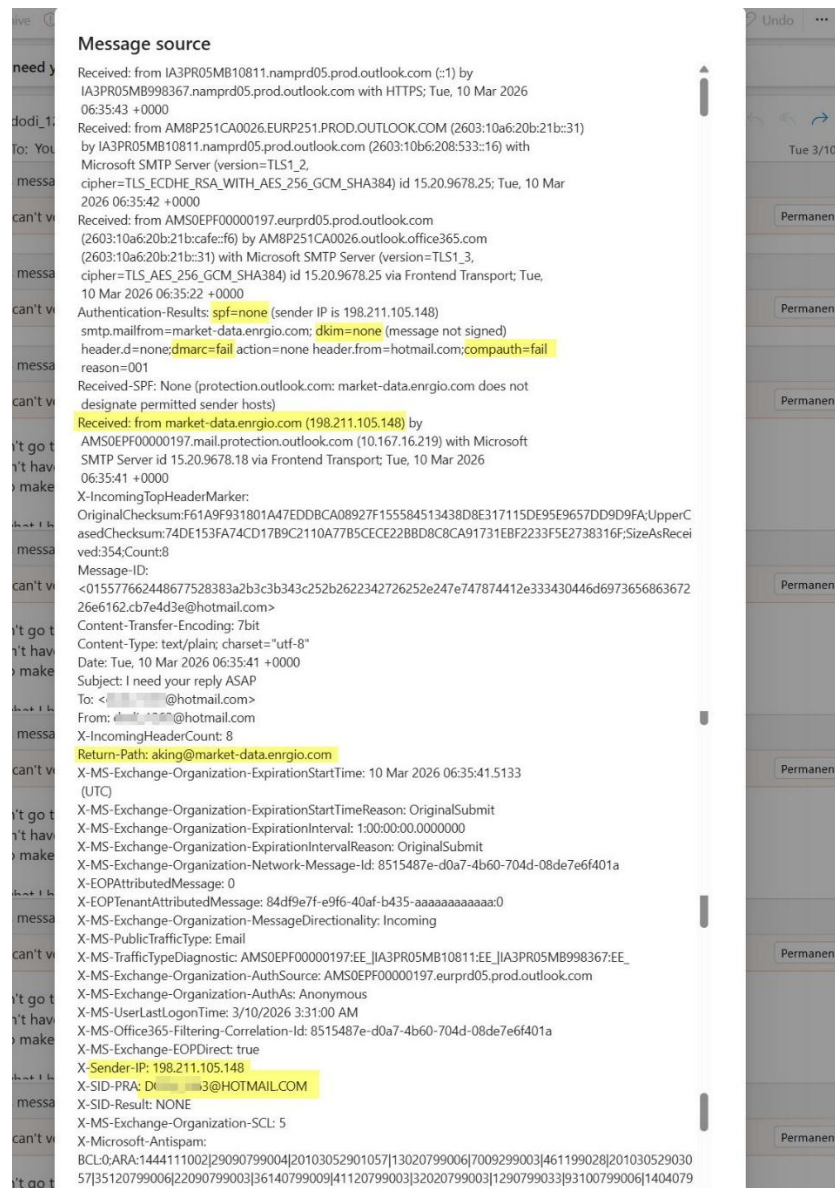
Appendix B — IP Infrastructure Analysis

This appendix contains:

- Header of emails (2-5)
- IP intelligence lookup for email (2-5)
- AbuseIPDB results for email (2-5)

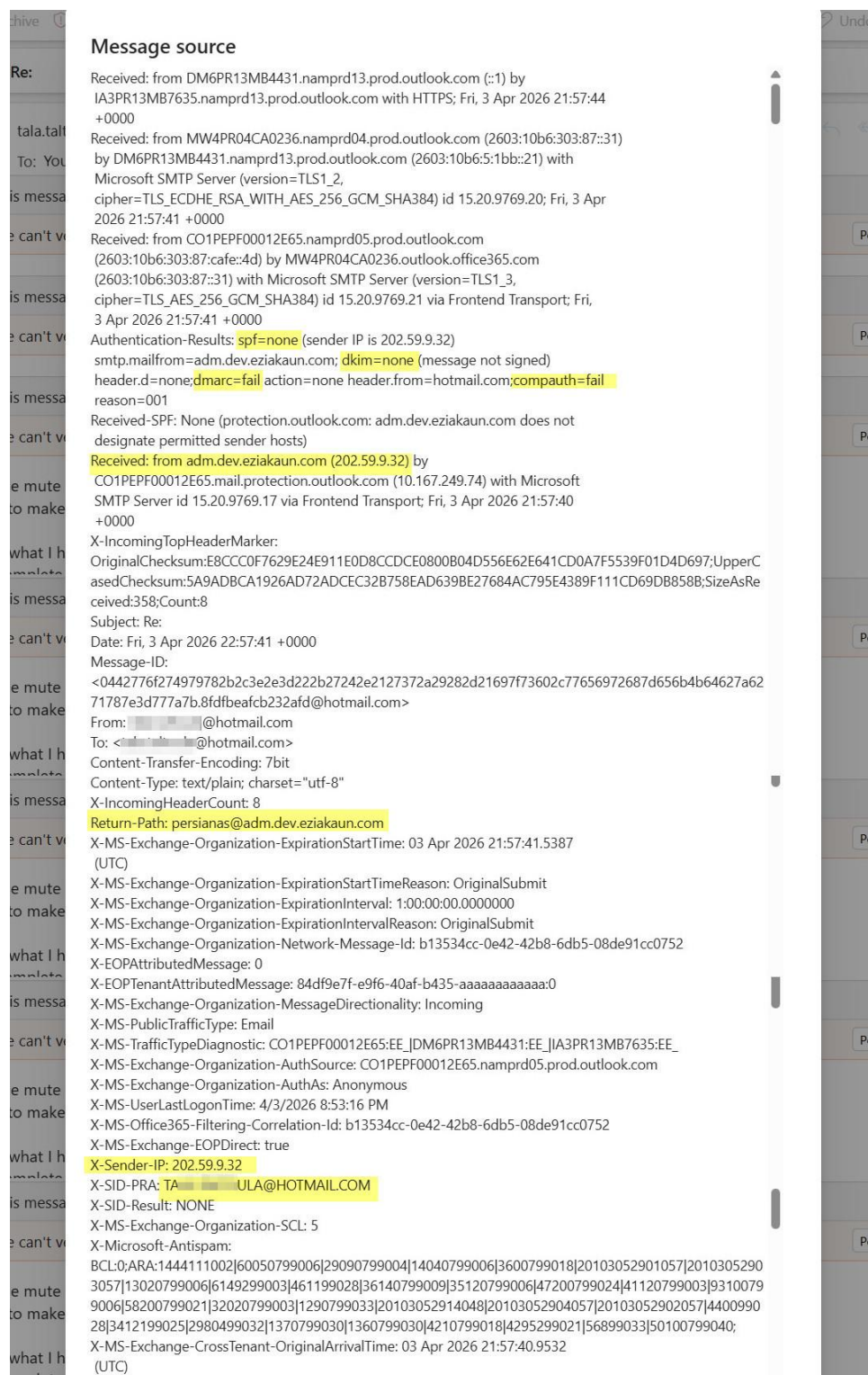
Linked to section: 4.1 Overview

Figure B1: Email 2 content



Extract of email header showing originating IP, SMTP path, and authentication results used for infrastructure analysis

Figure B2: Email 3 content



Extract of email header showing originating IP, SMTP path, and authentication results used for infrastructure analysis

Figure B3: Email 4 content

Message source

Received: from SA0PR07MB7658.namprd07.prod.outlook.com (::1) by SA5PR07MB11723.namprd07.prod.outlook.com with HTTPS; Sat, 11 Apr 2026 01:23:06 +0000

Received: from CH5PR04CA0001.namprd04.prod.outlook.com (2603:10b6:610:1f4::6) by SA0PR07MB7658.namprd07.prod.outlook.com (2603:10b6:806:be::8) with Microsoft SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.9769.17; Sat, 11 Apr 2026 01:23:05 +0000

Received: from DS3PEPF000099E1.namprd04.prod.outlook.com (2603:10b6:610:1f4:cafe::e8) by CH5PR04CA0001.outlook.office365.com (2603:10b6:610:1f4::6) with Microsoft SMTP Server (version=TLS1_3, cipher=TLS_AES_256_GCM_SHA384) id 15.20.9769.43 via Frontend Transport; Sat, 11 Apr 2026 01:23:05 +0000

Authentication-Results: **spf=none** (sender IP is 152.42.239.74) smtp.mailfrom=**cggl.com**; **dkim=none** (message not signed) header.d=**none**; **dmARC=fail** action=none header.from=hotmail.com; **compauth=fail** reason=001

Received-SPF: None (protection.outlook.com: cggl.com does not designate permitted sender hosts)

Received: from localhost (152.42.239.74) by DS3PEPF000099E1.mail.protection.outlook.com (10.167.17.196) with Microsoft SMTP Server id 15.20.9769.17 via Frontend Transport; Sat, 11 Apr 2026 01:23:03 +0000

X-IncomingTopHeaderMarker:
OriginalChecksum:725C3116632EDC49931466E7DDDBFEFF97E6288695B08855889E83B9D4619F0;UpperCaseChecksum:156E4A60CB9CB9ED0FE28886B169F1CCB85BF40CCF48D52C7B18F7BD5D327733;SizeAsReceived:365;Count:8

From: abc [redacted]mad@hotmail.com

Date: Sat, 11 Apr 2026 01:23:04 +0000

Subject: RE:

Content-Transfer-Encoding: base64

To: <[redacted]@hotmail.com>

Content-Type: text/plain; charset="utf-8"

Message-ID: <095a6b7f2c4c7e7d3020233324242c252b232a2f3c2f2e2d373f62666a735860646f6f536c66627175527b7b617b76717534787370.8b8d9e23d7ef@hotmail.com>

X-IncomingHeaderCount: 8

Return-Path: **cggl@cggl.com**

X-MS-Exchange-Organization-ExpirationStartTime: 11 Apr 2026 01:23:04.9686 (UTC)

X-MS-Exchange-Organization-ExpirationStartTimeReason: OriginalSubmit

X-MS-Exchange-Organization-ExpirationInterval: 1:00:00:00.0000000

X-MS-Exchange-Organization-ExpirationIntervalReason: OriginalSubmit

X-MS-Exchange-Organization-Network-Message-Id: af1c0a50-d1b2-4906-d8fc-08de9768e18c

X-EOPAttributedMessage: 0

X-EOPTenantAttributedMessage: 84df9e7f-e9f6-40af-b435-aaaaaaaaaaaaa:0

X-MS-Exchange-Organization-MessageDirectionality: Incoming

X-MS-PublicTrafficType: Email

X-MS-TrafficTypeDiagnostic: DS3PEPF000099E1:EE_[SA0PR07MB7658:EE_[SA5PR07MB11723:EE_

X-MS-Exchange-Organization-AuthSource: DS3PEPF000099E1.namprd04.prod.outlook.com

X-MS-Exchange-Organization-AuthAs: Anonymous

X-MS-UserLastLogonTime: 4/10/2026 10:02:03 PM

X-MS-Office365-Filtering-Correlation-Id: af1c0a50-d1b2-4906-d8fc-08de9768e18c

X-MS-Exchange-EOPDirect: true

X-Sender-IP: **152.42.239.74**

X-SID-PRA: [redacted]@HOTMAIL.COM

X-SID-Result: NONE

X-MS-Exchange-Organization-SCL: 5

X-Microsoft-Antispam:
BCL:0;ARA:1444111002j6149299003j41010799006j84100799006j29090799004j9029299007j461199028j1302n799006j6005n799006j93100799006j36100799006j35120799006j14000799006j1120799003j1290799003j32

Extract of email header showing originating IP, SMTP path, and authentication results used for infrastructure analysis

Figure B4: Email 5 content

Message source

Received: from PH8PR07MB995354.namprd07.prod.outlook.com (::1) by SA5PR07MB11723.namprd07.prod.outlook.com with HTTPS; Fri, 27 Mar 2026 13:58:13 +0000

Received: from BY5PR03CA0009.namprd03.prod.outlook.com (2603:10b6:a03:1e0::19) by PH8PR07MB995354.namprd07.prod.outlook.com (2603:10b6:510:3a7::15) with Microsoft SMTP Server (version=TLS1_2, cipher=TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384) id 15.20.9745.23; Fri, 27 Mar 2026 13:58:13 +0000

Received: from SJ1PEPF000026C6.namprd04.prod.outlook.com (2603:10b6:a03:1e0:cafe::eb) by BY5PR03CA0009.outlook.office365.com (2603:10b6:a03:1e0:19) with Microsoft SMTP Server (version=TLS1_3, cipher=TLS_AES_256_GCM_SHA384) id 15.20.9723.33 via Frontend Transport; Fri, 27 Mar 2026 13:58:12 +0000

Authentication-Results: **spf=none** (sender IP is 45.79.252.40) smtp.mailfrom=casainterativa.com; **dkim=none** (message not signed) header.d=none;**dmARC=fail** action=none header.from=hotmail.com;**compauth=fail** reason=001

Received-SPF: None (protection.outlook.com: casainterativa.com does not designate permitted sender hosts)

Received: from 45-79-252-40.ip.linodeusercontent.com (45.79.252.40) by SJ1PEPF000026C6.mail.protection.outlook.com (10.167.244.103) with Microsoft SMTP Server id 15.20.9745.21 via Frontend Transport; Fri, 27 Mar 2026 13:58:12 +0000

X-IncomingTopHeaderMarker:
OriginalChecksum:C54B936890F936406259B32A0B02385840B28A044B37BE805DED10378040AB6F;UpperCaseChecksum:35D87C1D74D24EB686D476F8195DC4F9AAD9446A26B5501EF3691995D7A8C8B1;SizeAsReceived:408;Count:8

Content-Type: text/plain; charset="utf-8"

Message-ID: <1254617d355b766a39282c3c2c2d25343a3935372635383b3c366d6f617a4f797f76704a777f75787e5b74726a7241484e0d474a4b.0a96d8ca630d7128@hotmail.com>

Content-Transfer-Encoding: 7bit

Subject: Remember to pay your dues to avoid penalties

From: [REDACTED]@hotmail.com

To: [REDACTED]@hotmail.com

Date: Fri, 27 Mar 2026 13:58:12 +0000

X-IncomingHeaderCount: 8

Return-Path: janine@casainterativa.com

X-MS-Exchange-Organization-ExpirationStartTime: 27 Mar 2026 13:58:12.7821 (UTC)

X-MS-Exchange-Organization-ExpirationStartTimeReason: OriginalSubmit

X-MS-Exchange-Organization-ExpirationInterval: 1:00:00:00.0000000

X-MS-Exchange-Organization-ExpirationIntervalReason: OriginalSubmit

X-MS-Exchange-Organization-Network-Message-Id: 7c6f32d5-bcbc-4a7e-4b10-08de8c08e2ea

X-EOPAttributedMessage: 0

X-EOPTenantAttributedMessage: 84df9e7f-e9f6-40af-b435-aaaaaaaaaaaa:0

X-MS-Exchange-Organization-MessageDirectionality: Incoming

X-MS-PublicTrafficType: Email

X-MS-TrafficTypeDiagnostic: SJ1PEPF000026C6:EE_JPH8PR07MB995354:EE_JSA5PR07MB11723:EE_

X-MS-Exchange-Organization-AuthSource: SJ1PEPF000026C6.namprd04.prod.outlook.com

X-MS-Exchange-Organization-AuthAs: Anonymous

X-MS-UserLastLogonTime: 3/27/2026 1:38:06 PM

X-MS-Office365-Filtering-Correlation-Id: 7c6f32d5-bcbc-4a7e-4b10-08de8c08e2ea

X-MS-Exchange-EOPDirect: true

X-Sender-IP: 45.79.252.40

X-SID-PRA: [REDACTED]@HOTMAIL.COM

X-SID-Result: NONE

X-MS-Exchange-Organization-SCL: 5

X-Microsoft-Antispam: BCL:0;ARA:1444111002[461199028][29090799004][3069299007][58110799003][1290799033][32020799003][41120799003][13020799006][93100799006][35120799006][36140799009][60050799006][22090799003][6019299003][1

Extract of email header showing originating IP, SMTP path, and authentication results used for infrastructure analysis

Linked to section: 4.2 IP Infrastructure Analysis

Figure B5: Email 2 content

The screenshot displays the IPinfo website interface. At the top, the IP address 198.211.105.148 is entered in the search bar. The navigation menu includes links for Products, Data, Why IPinfo?, Pricing, Resources, Docs, Login, and a Sign up button. Below the search bar, a breadcrumb trail shows the hierarchy: All IP Ranges > 198.0.0.0/8 > 198.211.0.0/16 > 198.211.105.0/24 > 198.211.105.148. The IP address 198.211.105.148 is prominently displayed in yellow, with icons for a folder, star, and share. Below the IP address, a location pin icon indicates the location: North Bergen, New Jersey, US. To the right of the location are icons for Cloud, Hosting, SSH, and Webserver. A blue button labeled 'Summary' is visible on the left sidebar. A green banner at the top right of the main content area reads: 'Need more data or want to access it via API or data downloads? Sign up to get free access' with a 'Sign up for free' button. The main content area is divided into two sections: 'Summary' and 'IP Geolocation'. The 'Summary' section contains a table with the following data: ASN (AS14061 - DigitalOcean, LLC), Hostname (No Hostname), Range (198.211.96.0/20), Company (DigitalOcean, LLC), Hosted domains (0), Privacy (True), Anycast (False), ASN type (Hosting), and Abuse contact (abuse@digitalocean.com). The 'IP Geolocation' section contains a table with the following data: City (North Bergen), State (New Jersey), Country (United States), Postal (07047), Local time (02:52 PM, Sunday, May 03, 2026), Timezone (America/New_York), and Coordinates (40.8043 N, 74.0121 W). To the right of the table is a map showing the location of North Bergen, New Jersey, with a red pin and the coordinates 40.8043 N, 74.0121 W. Below the map is a link to 'IP Geolocation API & Database'. At the bottom of the page, there is a footer with the text: 'Useful for Web Personalization, and Financial Technology'.

IPinfo 198.211.105.148 Products Data Why IPinfo? Pricing Resources Docs Login Sign up

All IP Ranges > 198.0.0.0/8 > 198.211.0.0/16 > 198.211.105.0/24 > 198.211.105.148

198.211.105.148 📁 ☆ Share

📍 North Bergen, New Jersey, US 🌐 Cloud 🖥️ Hosting 🔑 SSH 🖨️ Webserver

Summary >

Need more data or want to access it via API or data downloads? Sign up to get free access Sign up for free >

Summary

ASN	AS14061 - DigitalOcean, LLC
Hostname	No Hostname
Range	198.211.96.0/20
Company	DigitalOcean, LLC
Hosted domains	0
Privacy	✓ True
Anycast	✗ False
ASN type	Hosting
Abuse contact	abuse@digitalocean.com

IP Geolocation

City	North Bergen
State	New Jersey
Country	🇺🇸 United States
Postal	07047
Local time	02:52 PM, Sunday, May 03, 2026
Timezone	America/New_York
Coordinates	40.8043 N, 74.0121 W

IP Geolocation data

IP geolocation lookup is the identification of an IP address' geographic location in the real world.

Useful for Web Personalization, and Financial Technology

[IP Geolocation API & Database >](#)

IP intelligence lookup shows cloud-hosted infrastructure and geographic location associated with the sending server.

Figure B6: Email 3 content

[Products](#)
[Data](#)
[Why IPinfo?](#)
[Pricing](#)
[Resources](#)
[Docs](#)
[Login](#)
[Sign up](#)

[All IP Ranges](#)
[202.0.0.0/8](#)
[202.59.0.0/16](#)
[202.59.9.0/24](#)
[202.59.9.32](#)

202.59.9.32

[🔗](#)
[☆](#)
[Star](#)

Cyberjaya, Selangor, MY

Hosting

Summary

Geolocation
Privacy
ASN
Company
Abuse
ASN
Company
Abuse
ASN
Company
Abuse
ASN
Company
Abuse

Need more data or want to access it via API or data downloads? Sign up to get free access
Sign up for free

Summary

ASN	AS55720 - Gigabit Hosting Sdn Bhd
Hostname	No Hostname
Range	202.59.9.0/24
Company	Kon Jew Chung t/a Korx Solution
Hosted domains	0
Privacy	True
Anycast	False
ASN type	Hosting
Abuse contact	abuse@vpsmalaysia.com.my

IP Geolocation

City	Cyberjaya
State	Selangor
Country	Malaysia
Postal	63100
Local time	02:54 AM, Monday, May 04, 2026
Timezone	Asia/Kuala_Lumpur
Coordinates	2.9228 N, 101.6572 E

IP Geolocation data

IP geolocation lookup is the identification of an IP address' geographic location in the real world.

Useful for Web Personalization, and Financial Technology

[IP Geolocation API & Database](#)

IP intelligence lookup shows cloud-hosted infrastructure and geographic location associated with the sending server.

Figure B7: Email 4 content

IPinfo

152.42.239.74

Products

Data

Why IPinfo?

Pricing

Resources

Docs

Login

Sign up

All IP Ranges > 152.0.0.0/8 > 152.42.0.0/16 > 152.42.239.0/24 > 152.42.239.74

152.42.239.74

Star

Singapore, Singapore, SG

Cloud

Hosting

SSH

Webserver

Summary

Need more data or want to access it via API or data downloads? Sign up to get free access

Sign up for free

Geolocation

Privacy

ASN

Company

Abuse

ASN

Company

Abuse

ASN

Company

Abuse

ASN

Company

Abuse

Summary

ASNAS14061 - DigitalOcean, LLC

HostnameNo Hostname

Range152.42.224.0/20

CompanyDigitalOcean, LLC

Hosted domains0

PrivacyTrue

AnycastFalse

ASN typeHosting

Abuse contactabuse@digitalocean.com

IP Geolocation

CitySingapore

StateSingapore

CountrySingapore

Postal627753

Local time02:56 AM, Monday, May 04, 2026

TimezoneAsia/Singapore

Coordinates1.3215 N, 103.6957 E

Maps

1.3215 N, 103.6957 E

IP Geolocation data

IP geolocation lookup is the identification of an IP address' geographic location in the real world.

IP Geolocation API & Database

Useful for Web Personalization, and Financial Technology

IP intelligence lookup shows cloud-hosted infrastructure and geographic location associated with the sending server.

Figure B8: Email 5 content

IPinfo 45.79.252.40 Products Data Why IPinfo? Pricing Resources Docs Login Sign up

All IP Ranges > 45.0.0.0/8 > 45.79.0.0/16 > 45.79.252.0/24 > 45.79.252.40

45.79.252.40 📄 ☆ Star

📍 Atlanta, Georgia, US ☁ Cloud 🖨 Hosting 🔑 SSH 🖨 Webserver

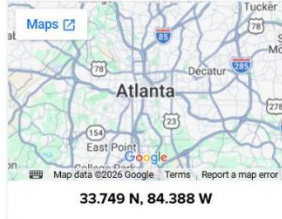
Need more data or want to access it via API or data downloads? Sign up to get free access Sign up for free

Summary

ASN	AS63949 - Akamai Connected Cloud
Hostname	45-79-252-40.ip.linodeusercontent.com
Range	45.79.252.0/22
Company	Linode
Hosted domains	0
Privacy	✓ True
Anycast	✗ False
ASN type	Hosting
Abuse contact	abuse@linode.com

IP Geolocation

City	Atlanta
State	Georgia
Country	🇺🇸 United States
Postal	30302
Local time	02:57 PM, Sunday, May 03, 2026
Timezone	America/New_York
Coordinates	33.749 N, 84.388 W



33.749 N, 84.388 W

IP Geolocation data

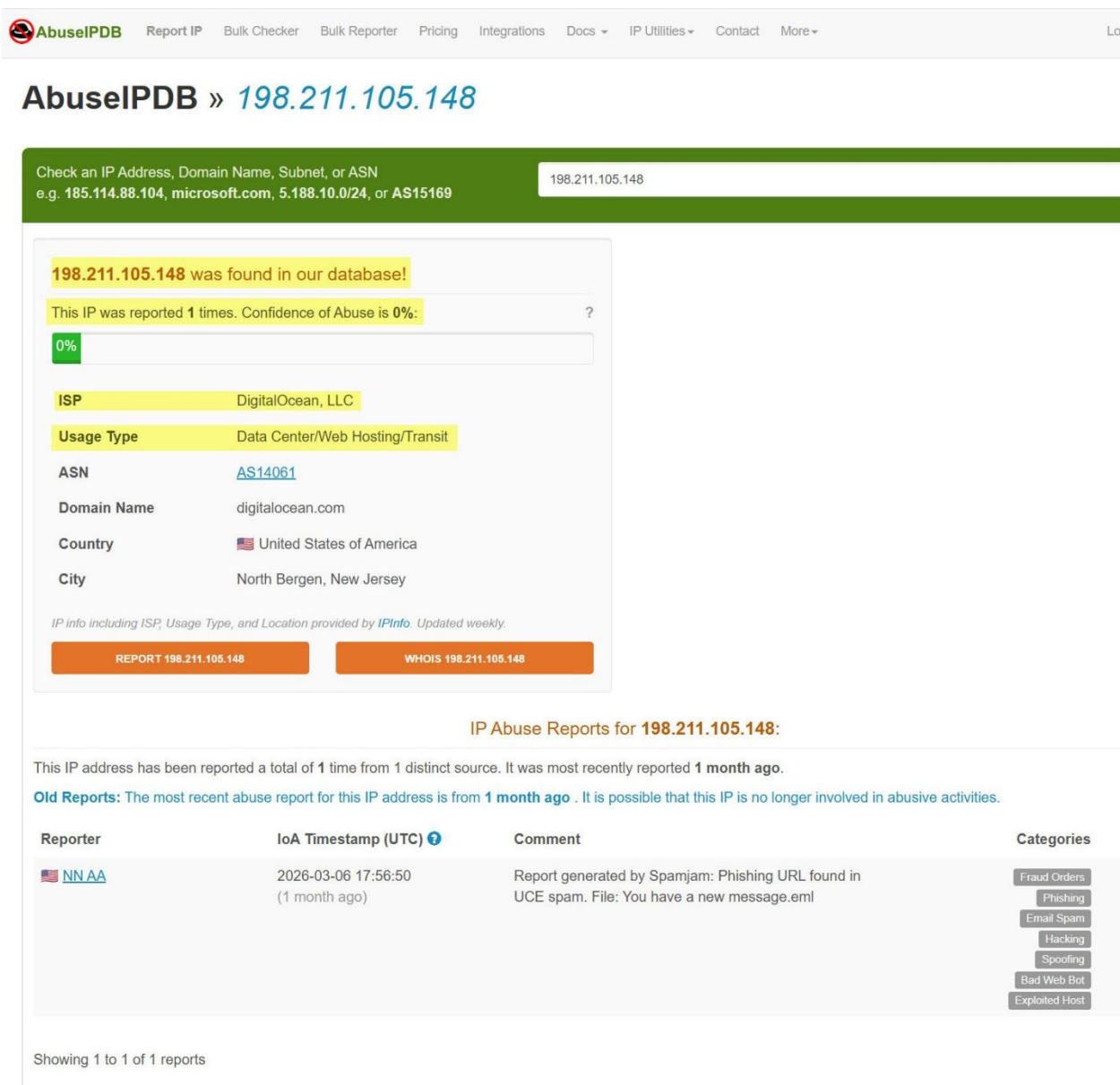
IP geolocation lookup is the identification of an IP address' geographic location in the real world.

Useful for Web Personalization, and Financial Technology

[IP Geolocation API & Database](#)

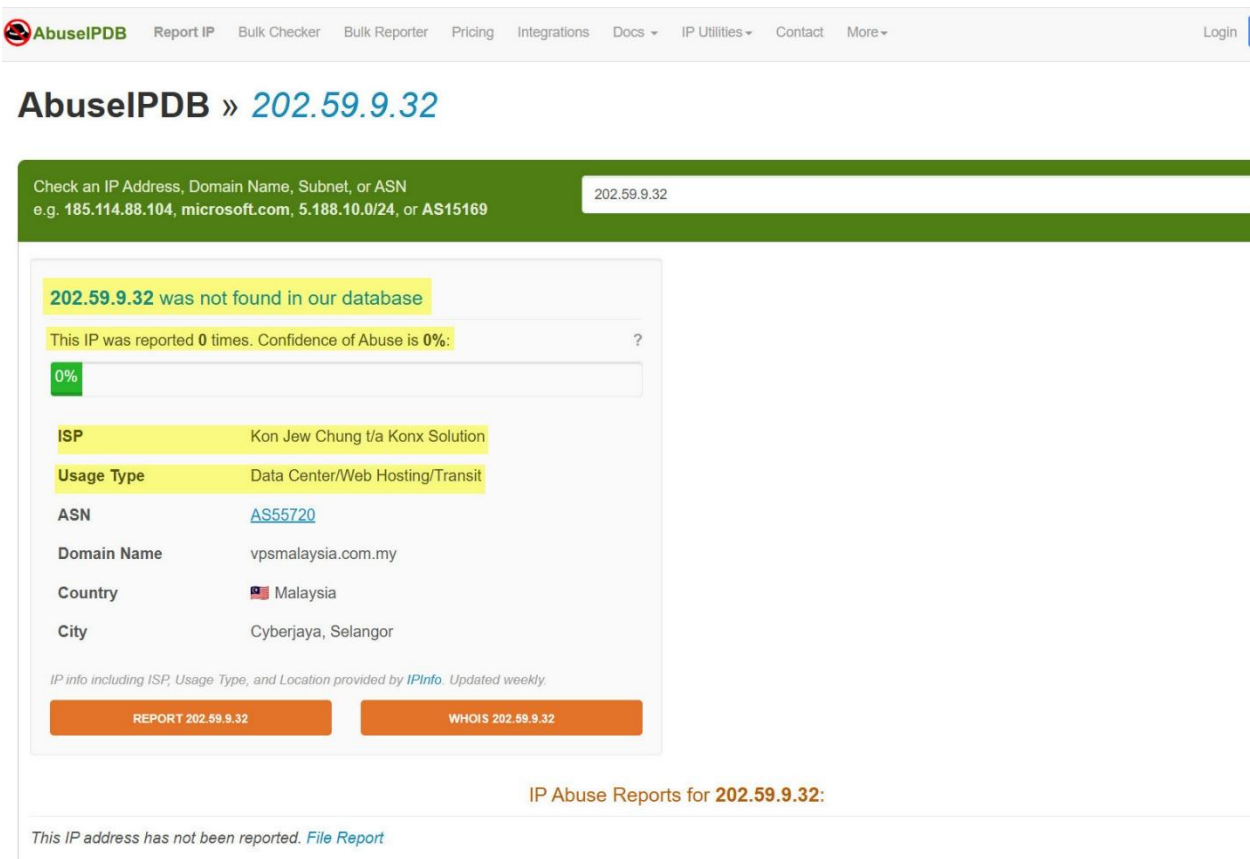
IP intelligence lookup shows cloud-hosted infrastructure and geographic location associated with the sending server.

Figure B9: Email 2 content



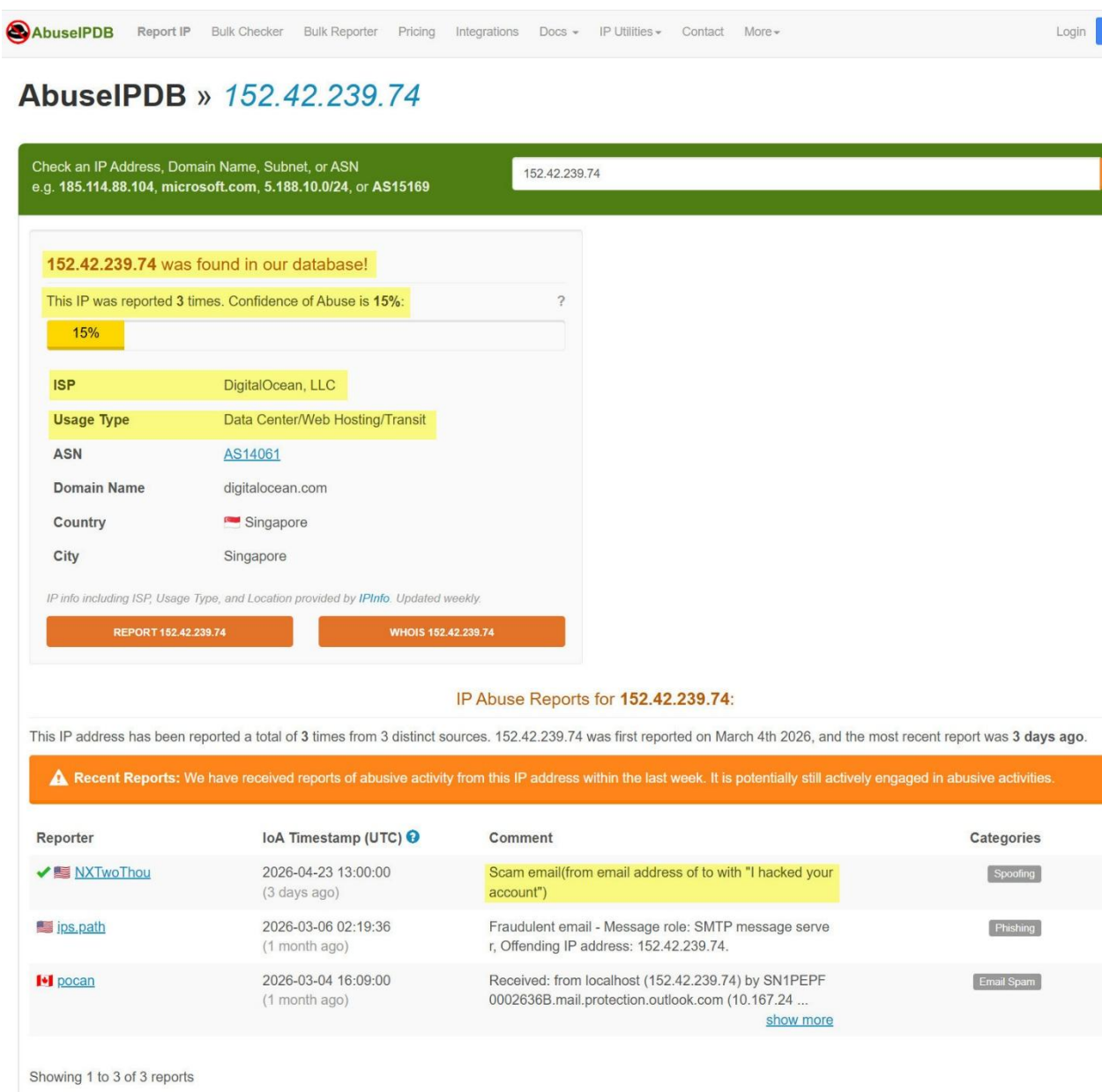
AbuseIPDB results showing historical abuse reports linked to extortion email activity and phishing behavior associated with the IP address.

Figure B10: Email 3 content



AbuseIPDB results showing historical abuse reports linked to extortion email activity and phishing behavior associated with the IP address.

Figure B11: Email 4 content



AbuseIPDB results showing historical abuse reports linked to extortion email activity and phishing behavior associated with the IP address.

Figure B12: Email 5 content

AbuseIPDB

[Report IP](#)[Bulk Checker](#)[Bulk Reporter](#)[Pricing](#)[Integrations](#)[Docs](#)[IP Utilities](#)[Contact](#)[More](#)

AbuseIPDB » 45.79.252.40

Check an IP Address, Domain Name, Subnet, or ASN
e.g. 185.114.88.104, microsoft.com, 5.188.10.0/24, or AS15169

45.79.252.40

45.79.252.40 was found in our database!

This IP was reported 1 times. Confidence of Abuse is 0%:

0%

ISP

Linode

Usage Type

Data Center/Web Hosting/Transit

ASN

[AS63949](#)

Hostname(s)

45-79-252-40.ip.linodeusercontent.com

Domain Name

linode.com

Country

United States of America

City

Atlanta, Georgia

IP info including ISP, Usage Type, and Location provided by [IPInfo](#). Updated weekly.

REPORT 45.79.252.40

WHOIS 45.79.252.40

IP Abuse Reports for 45.79.252.40:

This IP address has been reported a total of 1 time from 1 distinct source. It was most recently reported 1 year ago.

Old Reports: The most recent abuse report for this IP address is from 1 year ago . It is possible that this IP is no longer involved in abusive activities.

Reporter	IoA Timestamp (UTC) ?	Comment	Categories
✓ Anonymous	2024-05-17 09:01:34 (1 year ago)	Ports: 80,443; Direction: 0; Trigger: LF_CUSTOMTRIGGER	Brute-Force SSH

Showing 1 to 1 of 1 reports

AbuseIPDB results showing historical abuse reports linked to extortion email activity and phishing behavior associated with the IP address.

Appendix C: Domain Intelligence (WHOIS Analysis)

This appendix contains:

- WHOIS registration details
- Registrar and privacy protection usage
- Domain lifecycle indicators (registration/update/expiry)

Linked to section: 4.3 Domain and Infrastructure Configuration

Figure C1: Email 2 content

The screenshot displays a Whois domain lookup for **enrgio.com**. The interface includes a navigation bar with links for Domains, Hosting, Websites, Email, Security, Whois, and Deals. The domain name **enrgio.com** is entered in the search bar, and the results are updated 1 second ago.

Domain Information

Domain:	enrgio.com
Registered On:	2015-12-03
Expires On:	2030-12-03
Updated On:	2025-11-03
Status:	client transfer prohibited
Name Servers:	ns1.digitalocean.com ns2.digitalocean.com ns3.digitalocean.com

Registrar Information

Registrar:	Register.com - Network Solutions, LLC
IANA ID:	9
Abuse Email:	domain.operations@web.com
Abuse Phone:	+1.8777228662

Registrant Contact

Name:	PERFECT PRIVACY, LLC
Street:	5335 Gate Parkway care of Network Solutions PO Box 459
City:	Jacksonville
State:	FL
Postal Code:	32256
Country:	US
Phone:	+1.5707088622
Email:	vw7uz25p55t@networksolutionsprivateregistration.com

WHOIS domain lookup showing privacy-protected registration, registrar details, and anonymized ownership information, indicating potential identity obfuscation.

Figure C2: Email 3 content

Whois Identity for everyone Domains Hosting Websites Email Security Whois Deals Enter Dom

eziakaun.com

Updated 1 second ago

Domain Information

Domain:	eziakaun.com
Registered On:	2021-03-22
Expires On:	2027-03-22
Updated On:	2026-02-26
Status:	client transfer prohibited
Name Servers:	jake.ns.cloudflare.com meiling.ns.cloudflare.com

Registrar Information

Registrar:	HOSTINGER operations, UAB
IANA ID:	1636
Abuse Email:	abuse-tracker@hostinger.com
Abuse Phone:	+37064503378

Registrant Contact

Country:	LT
Email:	https://www.hostinger.com/whois?domain=eziakaun.com&view=contact

WHOIS domain lookup showing privacy-protected registration, registrar details, and anonymized ownership information, indicating potential identity obfuscation.

Figure C3: Email 4 content

Whois
Identity for everyone

Domains

Hosting

Websites

Email

Security

Whois

Deals

Enter Domain

cggl.com

Updated 1 second ago 



Domain Information

Domain:

cggl.com

Registered On:

2000-04-09

Expires On:

2034-04-09

Updated On:

2025-07-31

Status:

client delete prohibited
client transfer prohibited
client update prohibited

Name Servers:

ns1.dnsnameservice.com
ns2.dnsnameservice.com



Registrar Information

Registrar:

Tucows Domains Inc.

IANA ID:

69

Abuse Email:

domainabuse@tucows.com

Abuse Phone:

+1.4165350123



Registrant Contact

Name:

GOTW Hostmaster

Organization:

Get On The Web Limited

Street:

Fawley House
2 Regatta Place

City:

Bourne End

State:

Buckinghamshire

Postal Code:

SL8 5TD

Country:

GB

Phone:

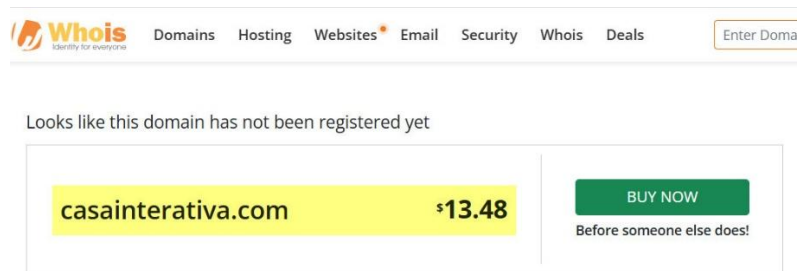
+44.1923902000

Email:

hostmaster@getontheweb.com

WHOIS domain lookup showing privacy-protected registration, registrar details, and anonymized ownership information, indicating potential identity obfuscation.

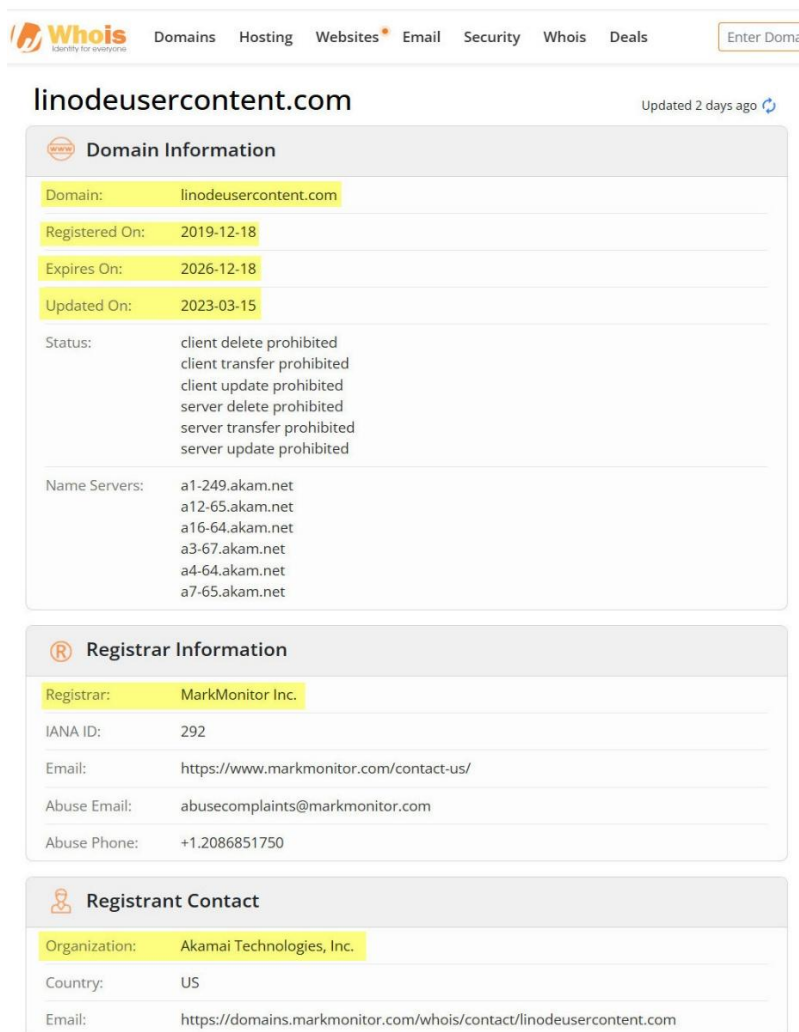
Figure C4: Email 5 content



The screenshot shows the Whois website interface. At the top, there is a navigation bar with links: Domains, Hosting, Websites, Email, Security, Whois, and Deals. A search bar on the right contains the text "Enter Doma". Below the navigation bar, a message states "Looks like this domain has not been registered yet". In the center, the domain "casainterativa.com" is displayed in a yellow box, followed by the price "\$13.48". To the right, there is a green "BUY NOW" button with the text "Before someone else does!" below it.

WHOIS domain lookup (return-path) showing privacy-protected registration, registrar details, and anonymized ownership information, indicating potential identity obfuscation.

Figure C5: Email 5 content



The screenshot shows the Whois website interface for the domain "linodeusercontent.com". The domain name is at the top, with a "Updated 2 days ago" link to its right. Below the domain name, there are three sections: "Domain Information", "Registrar Information", and "Registrant Contact".

Domain Information

Domain:	linodeusercontent.com
Registered On:	2019-12-18
Expires On:	2026-12-18
Updated On:	2023-03-15
Status:	client delete prohibited client transfer prohibited client update prohibited server delete prohibited server transfer prohibited server update prohibited
Name Servers:	a1-249.akam.net a12-65.akam.net a16-64.akam.net a3-67.akam.net a4-64.akam.net a7-65.akam.net

Registrar Information

Registrar:	MarkMonitor Inc.
IANA ID:	292
Email:	https://www.markmonitor.com/contact-us/
Abuse Email:	abusecomplaints@markmonitor.com
Abuse Phone:	+1.2086851750

Registrant Contact

Organization:	Akamai Technologies, Inc.
Country:	US
Email:	https://domains.markmonitor.com/whois/contact/linodeusercontent.com

WHOIS domain lookup showing privacy-protected registration, registrar details, and anonymized ownership information, indicating potential identity obfuscation.